

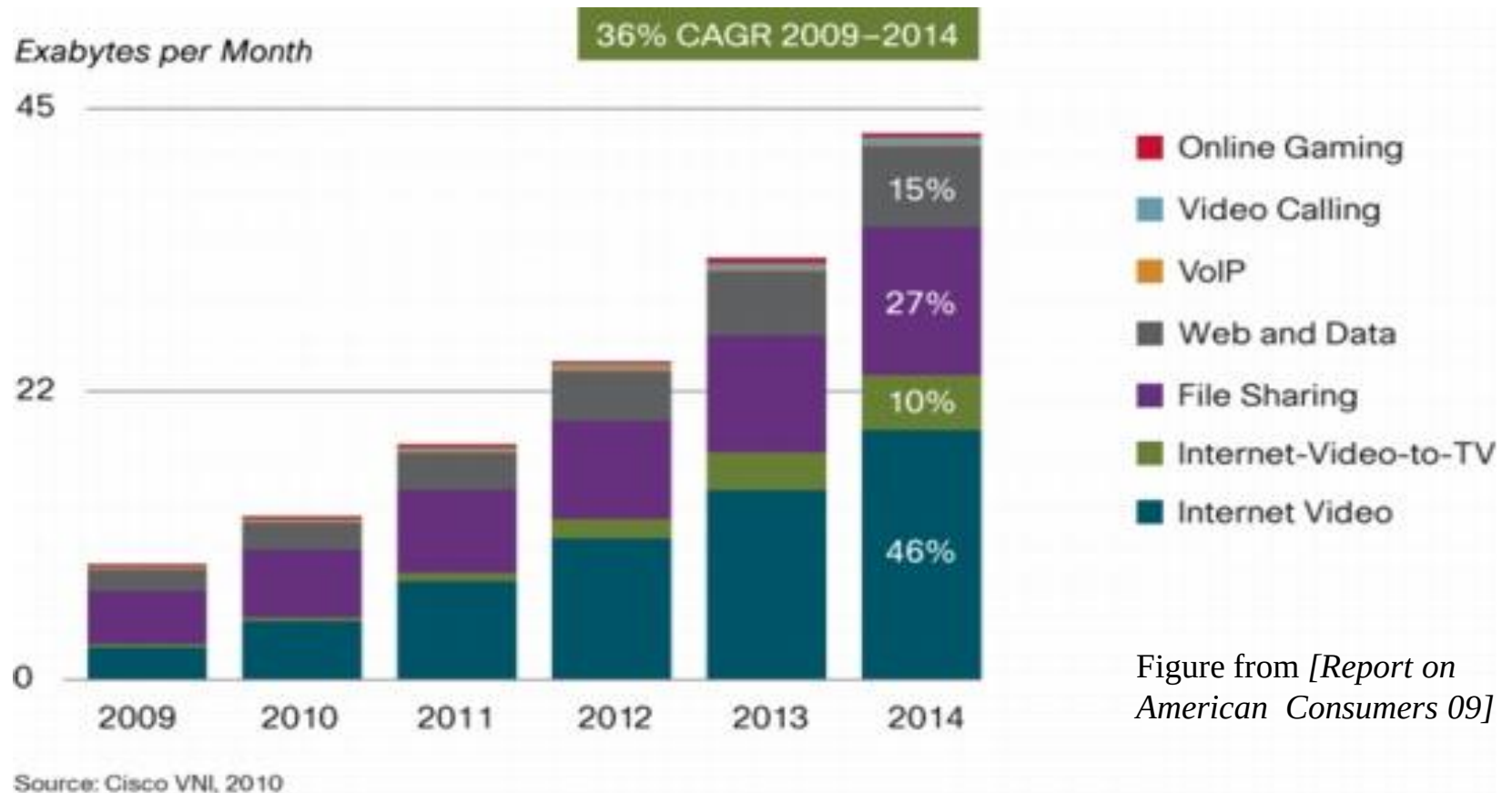
A Comprehensive Analysis and Parallelization of an Image Retrieval Algorithm

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Exploding Multimedia Data



Cisco VNI Global Consumer Internet Traffic Forecast

Multimedia Retrieval App.

Important to retrieve useful data

- ▶ E.g. medical imagery, video recommendation

Data-intensive and computing-intensive

Significant challenges for real-time retrieval

Multi-core Era Is Coming

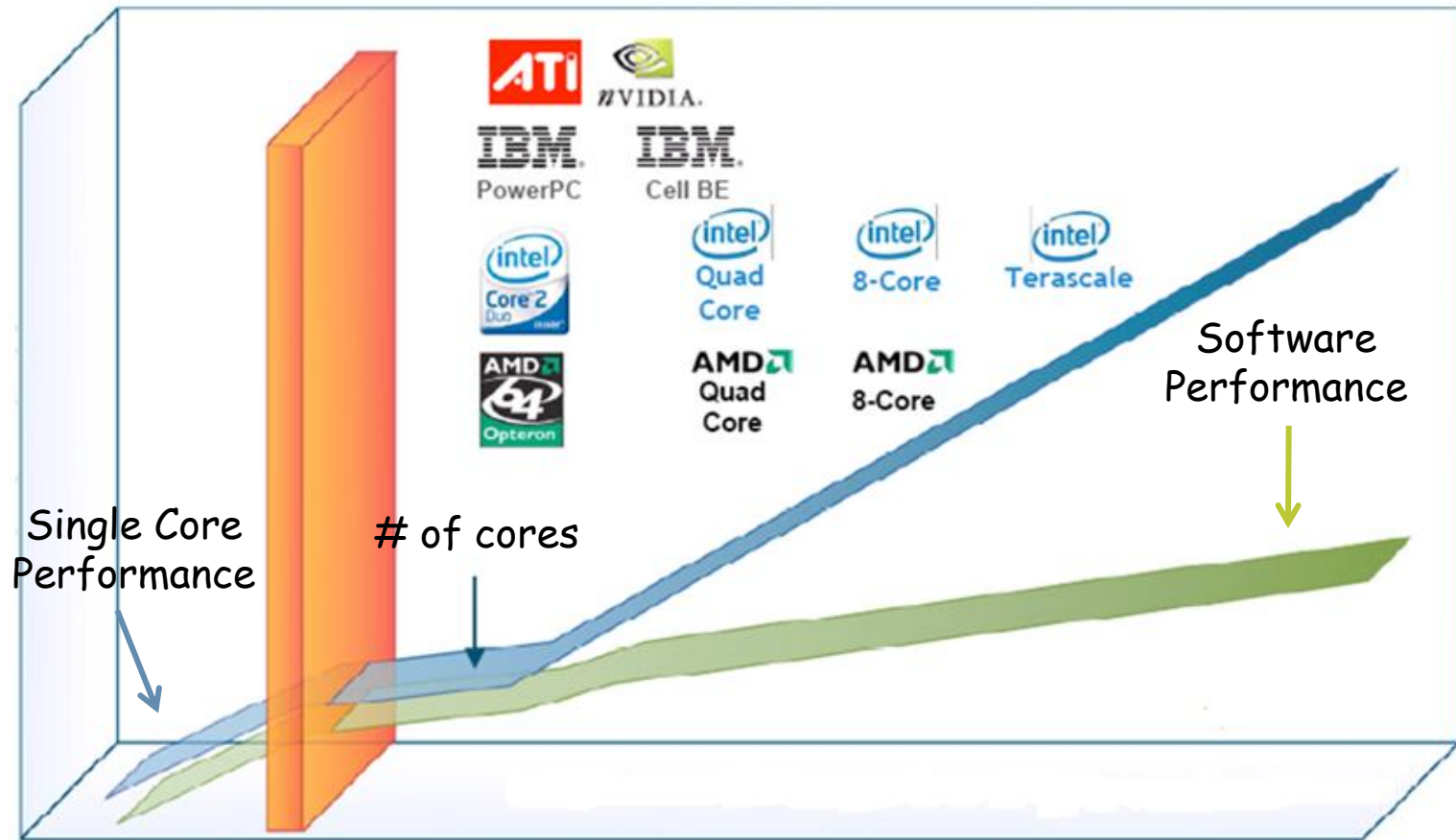


Figure from Michael MoCool's (Intel) many core slides

New Opportunities

Multi-core era needs parallelism

Need a comprehensive study on parallelism characteristics in multimedia retrieval

- ▶ To optimize them on current architectures
- ▶ To design future architectures for them

Image Retrieval

Image retrieval: also core of video retrieval



Image Retrieval

feature extraction

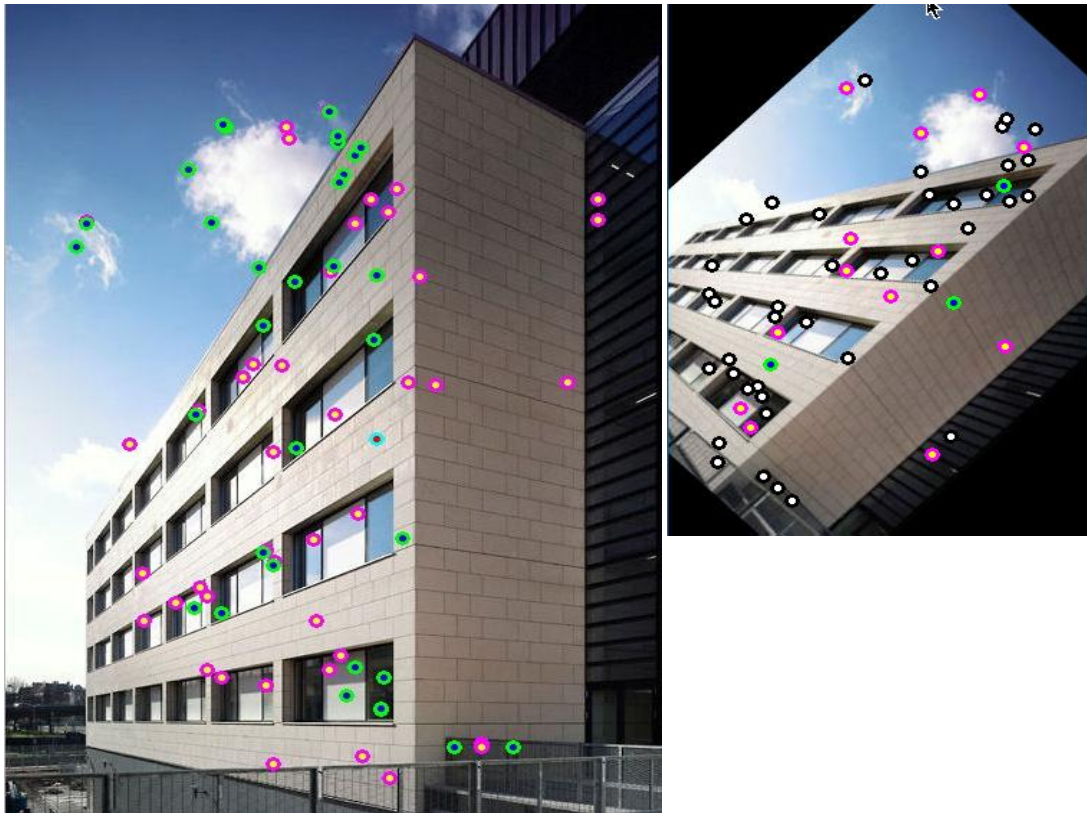


Image Retrieval

feature extraction + feature match



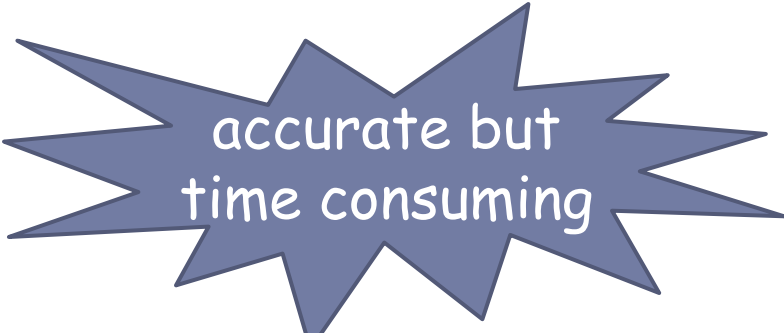
Image Retrieval (cont.)

Two classes of algorithms

- ▶ Global feature based
 - ▣ Color features
 - ▣ Texture features
- ▶ Local feature based
 - ▣ Shape context
 - ▣ SIFT Features
 - ▣ SURF Features



~60% precision
[Wan 08]



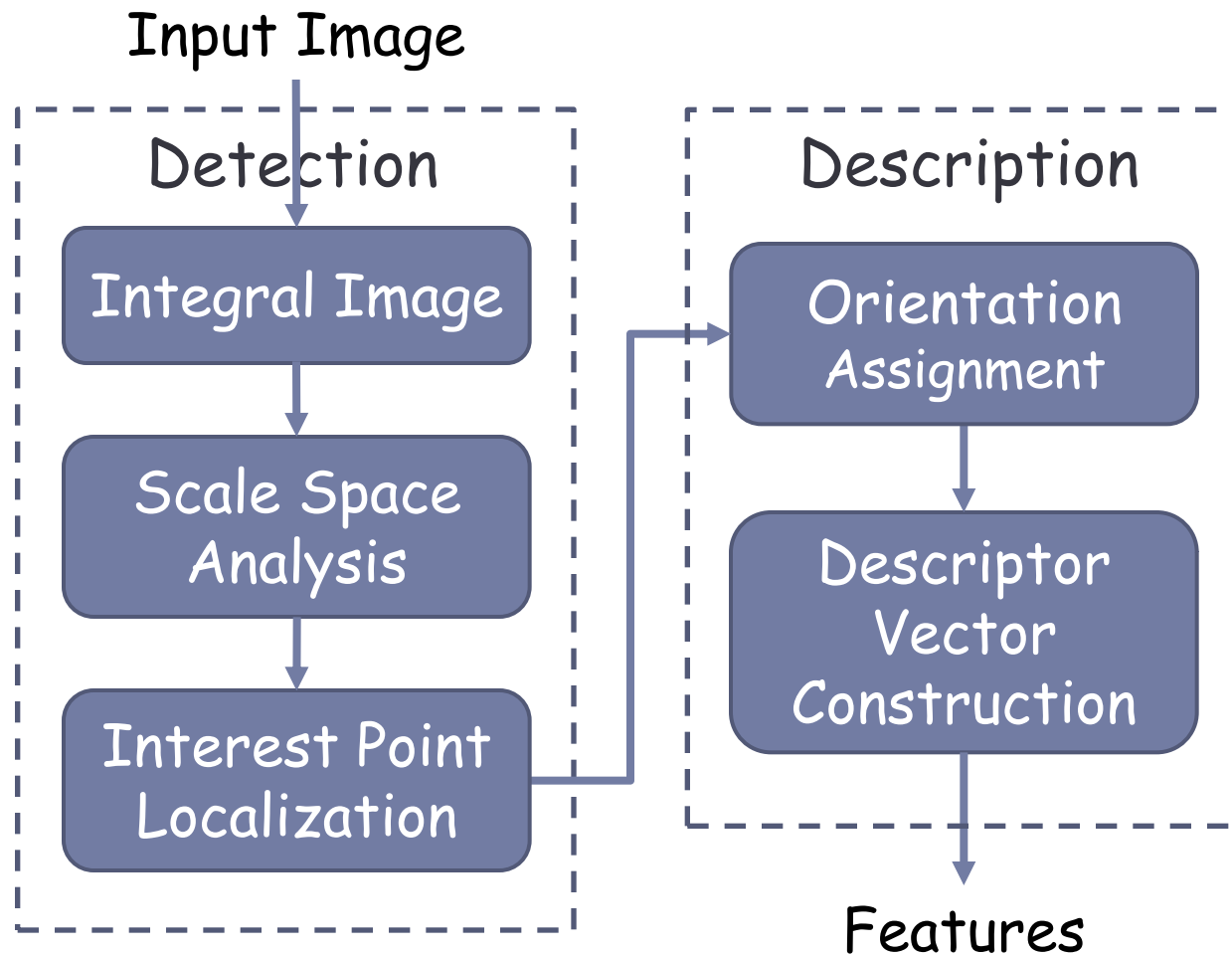
accurate but
time consuming



robust & appealing: *insensitive to
scale and rotation transformation*

[Mikolajczyk 05, Bauer 07]

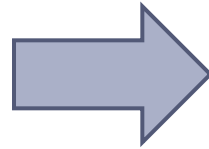
SURF Overview



Integral Image



Input Image



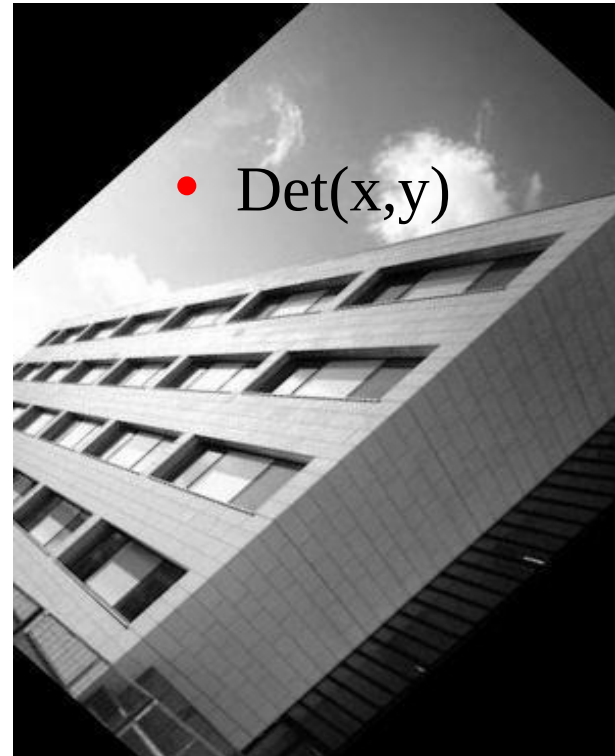
Integral Image

Scale Space Analysis

Hessian Matrix
for (x,y) [Bay 06] $\begin{pmatrix} D_{xx} & D_{xy} \\ D_{xy} & D_{yy} \end{pmatrix}$



...



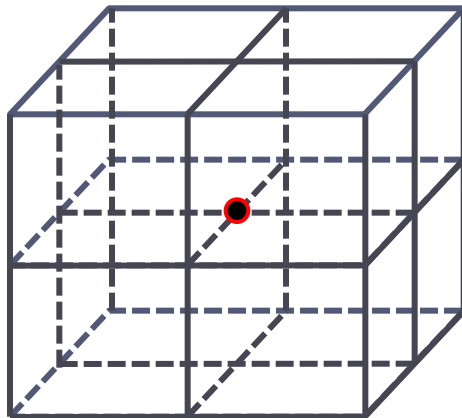
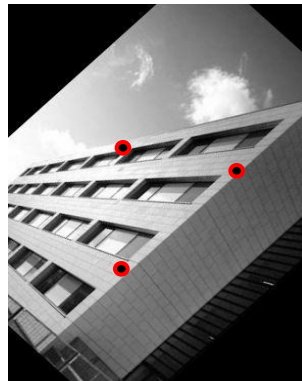
insensitive to scale
transformation

$$\text{Det}(x,y) = D_{xx} * D_{yy} - 0.81 * D_{xy} * D_{xy}$$

Interest Point Localization



...



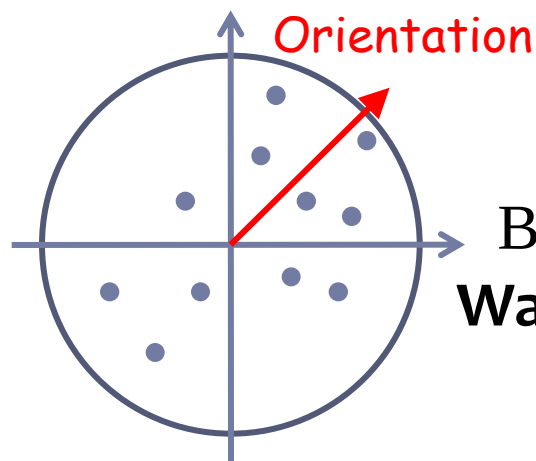
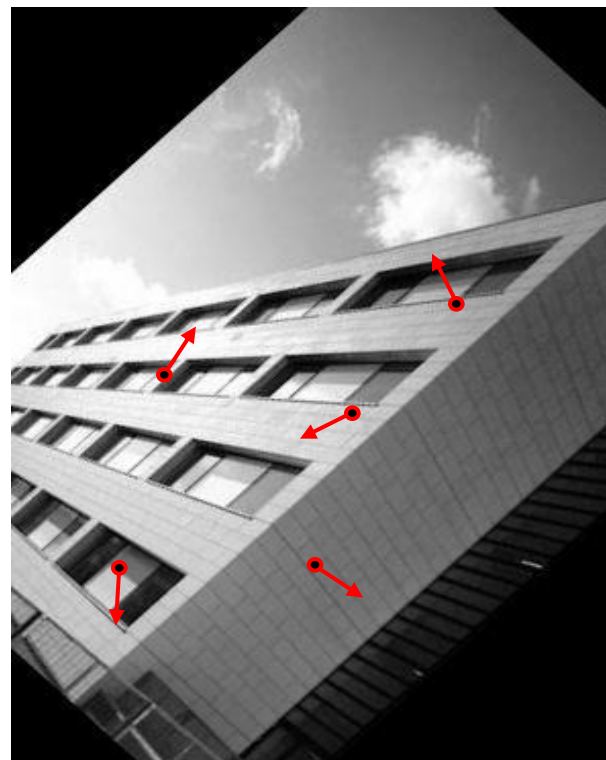
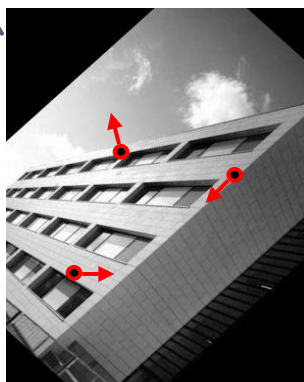
Point with
max det value

Orientation Assignment

insensitive to rotation
transformation

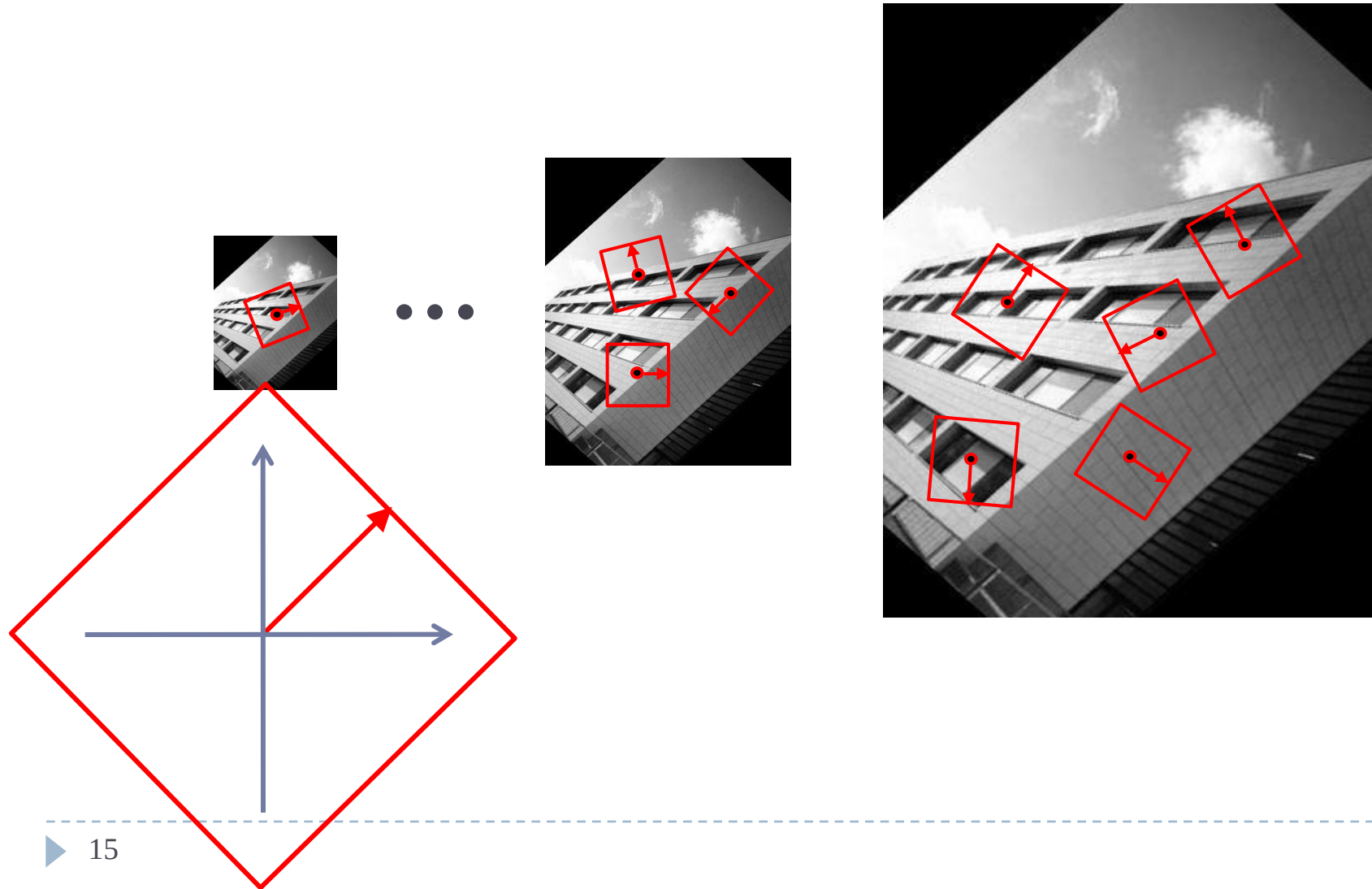


...



Based on **Haar
Wavelet** [*Bay 06*]

Descriptor Vector Construction

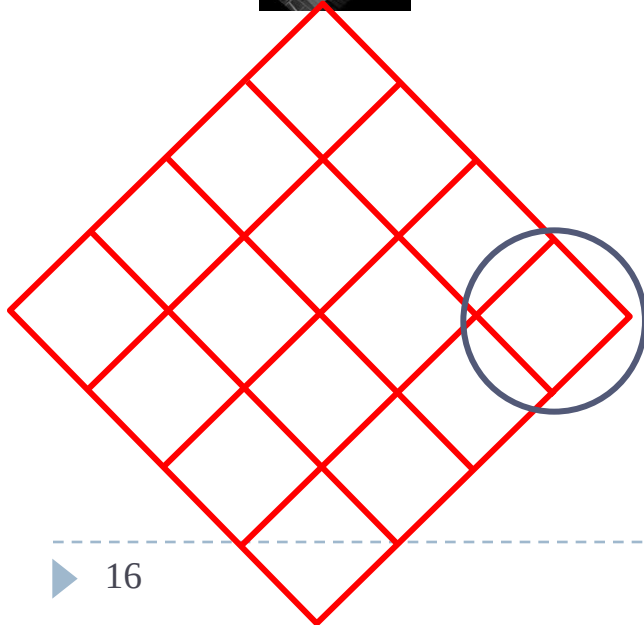


Descriptor Vector Construction

64-dimension feature vector

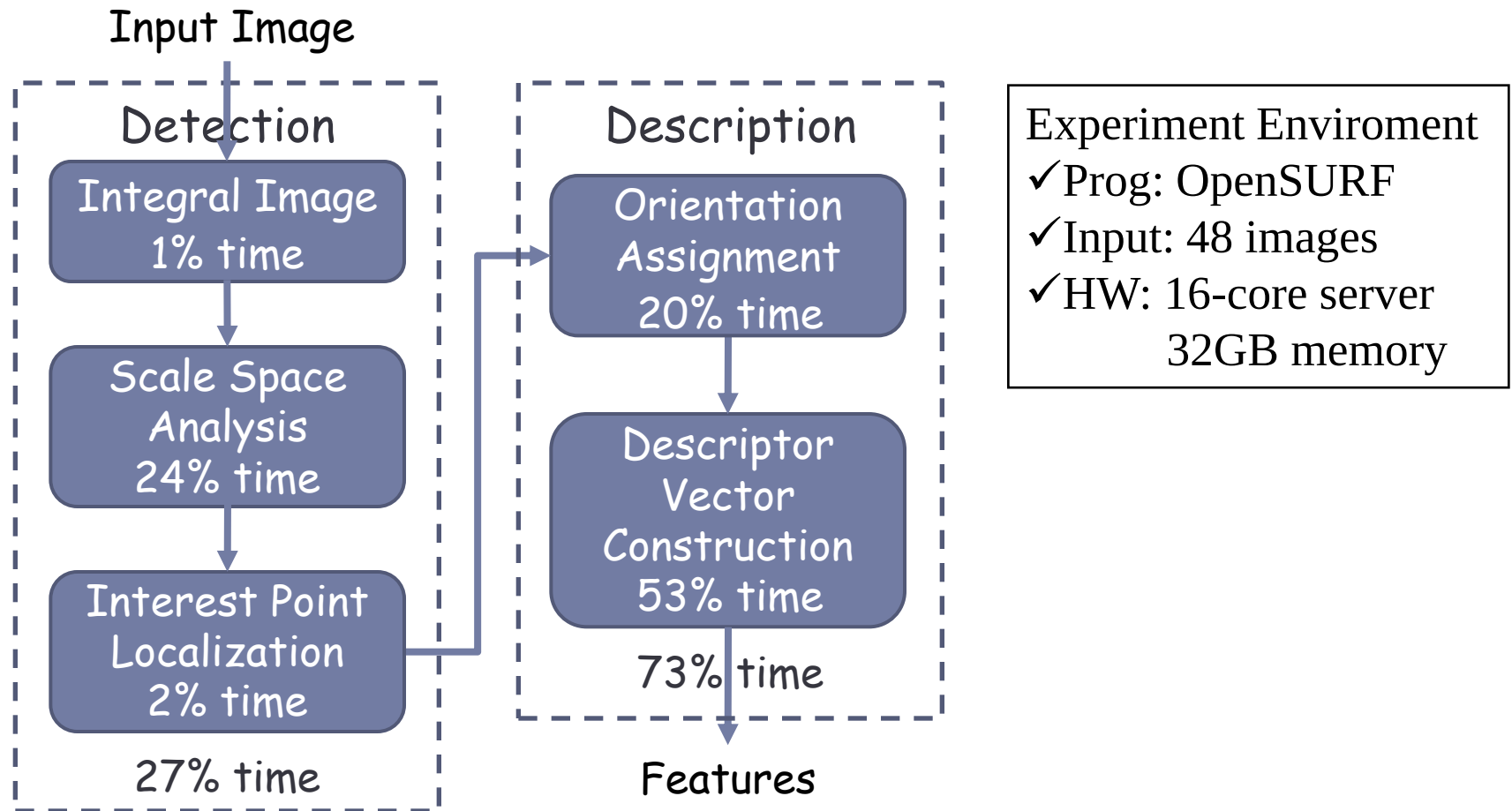


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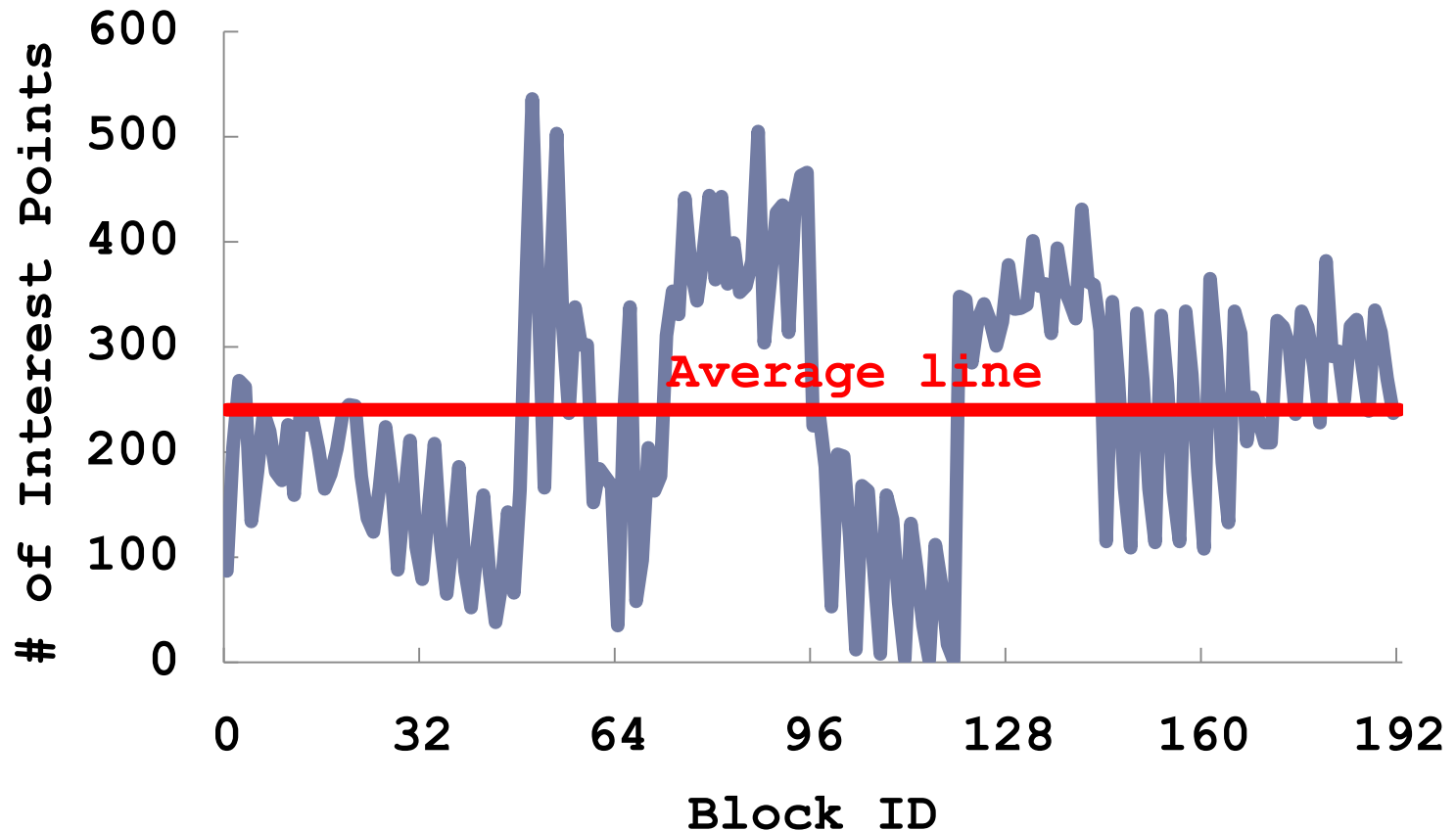
4-dimension vector calculated
based on **Haar Wavelet**

Execution Profile of SURF



Interest Points Distribution

Imbalanced distribution for images/blocks



Parallel Analysis

Pipeline Parallelism

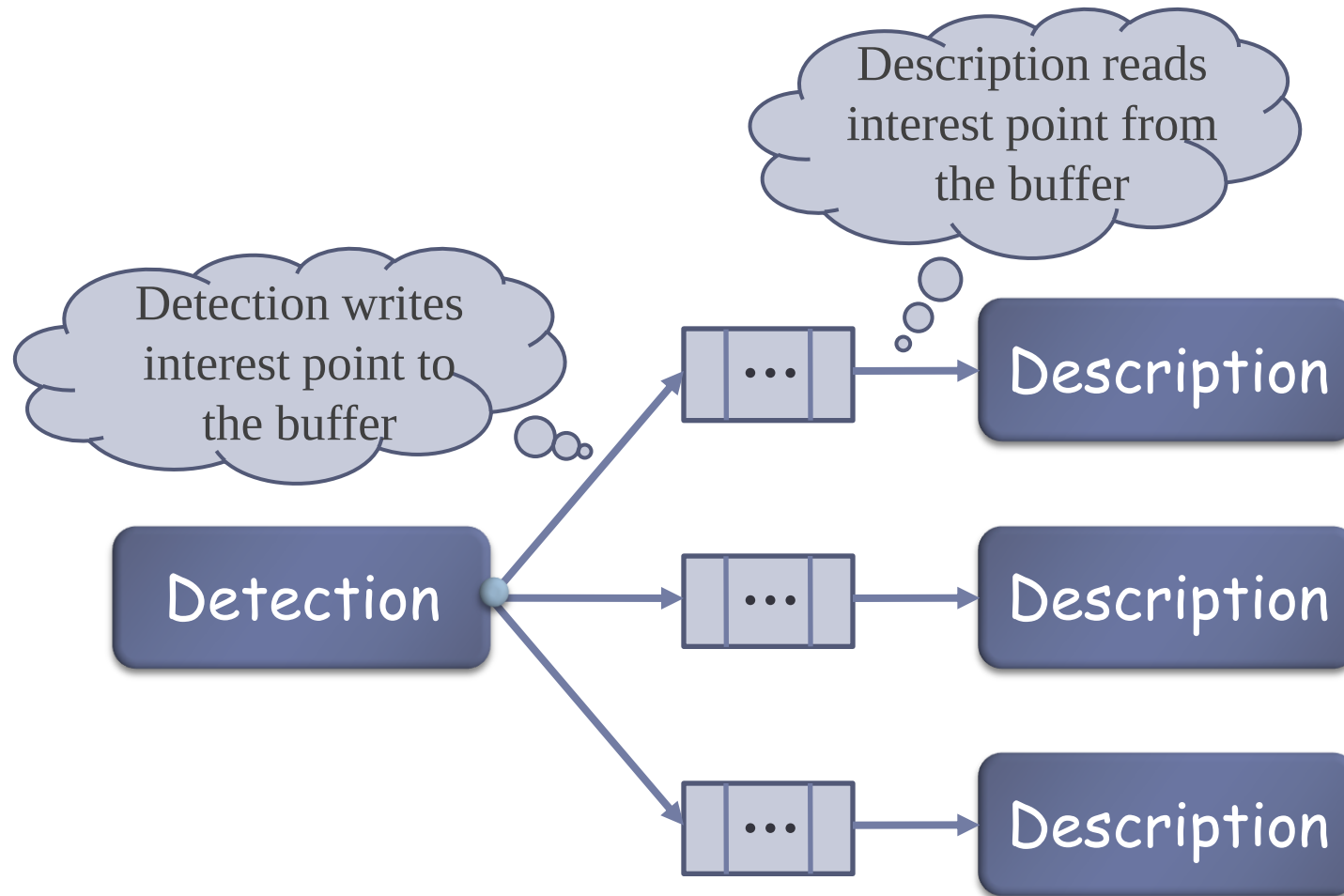
Task Parallelism

- ▶ Scale-level Parallelism
- ▶ Block-level Parallelism

Combination of Different Parallelism

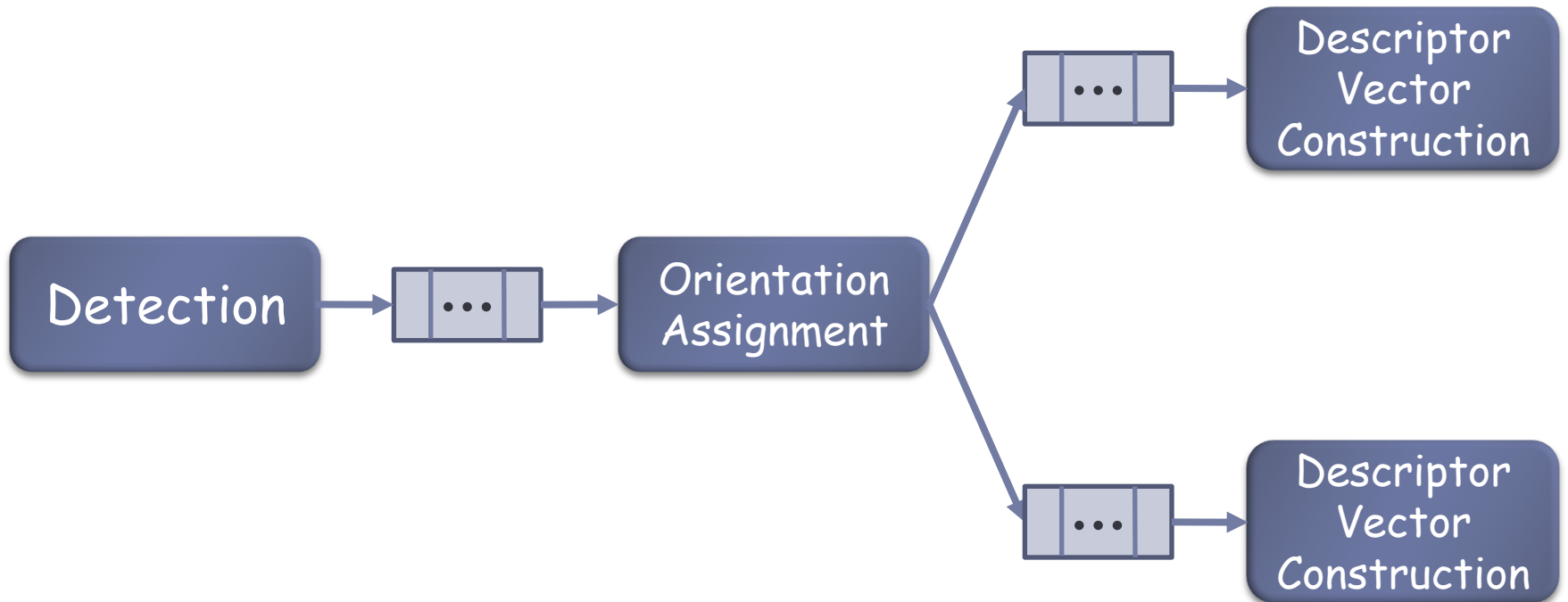
- ▶ Combination of SIMD and Other Parallelism
- ▶ Combination of Pipeline and Task Parallelism

2-stage Pipeline



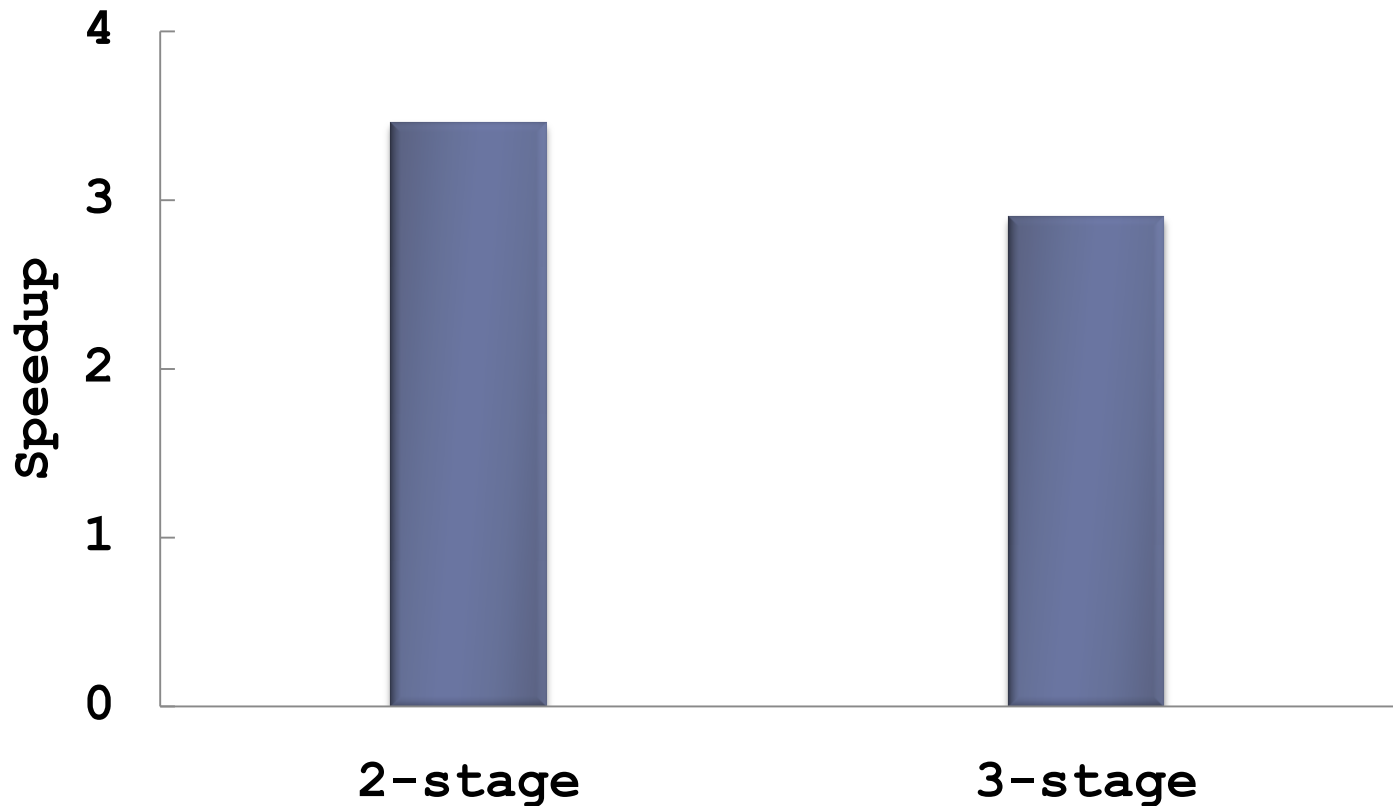
3-stage Pipeline

Further divide Description into two stages



Results of Pipeline Parallelism

Pipeline parallelism does not scale



Parallel Analysis

Pipeline Parallelism

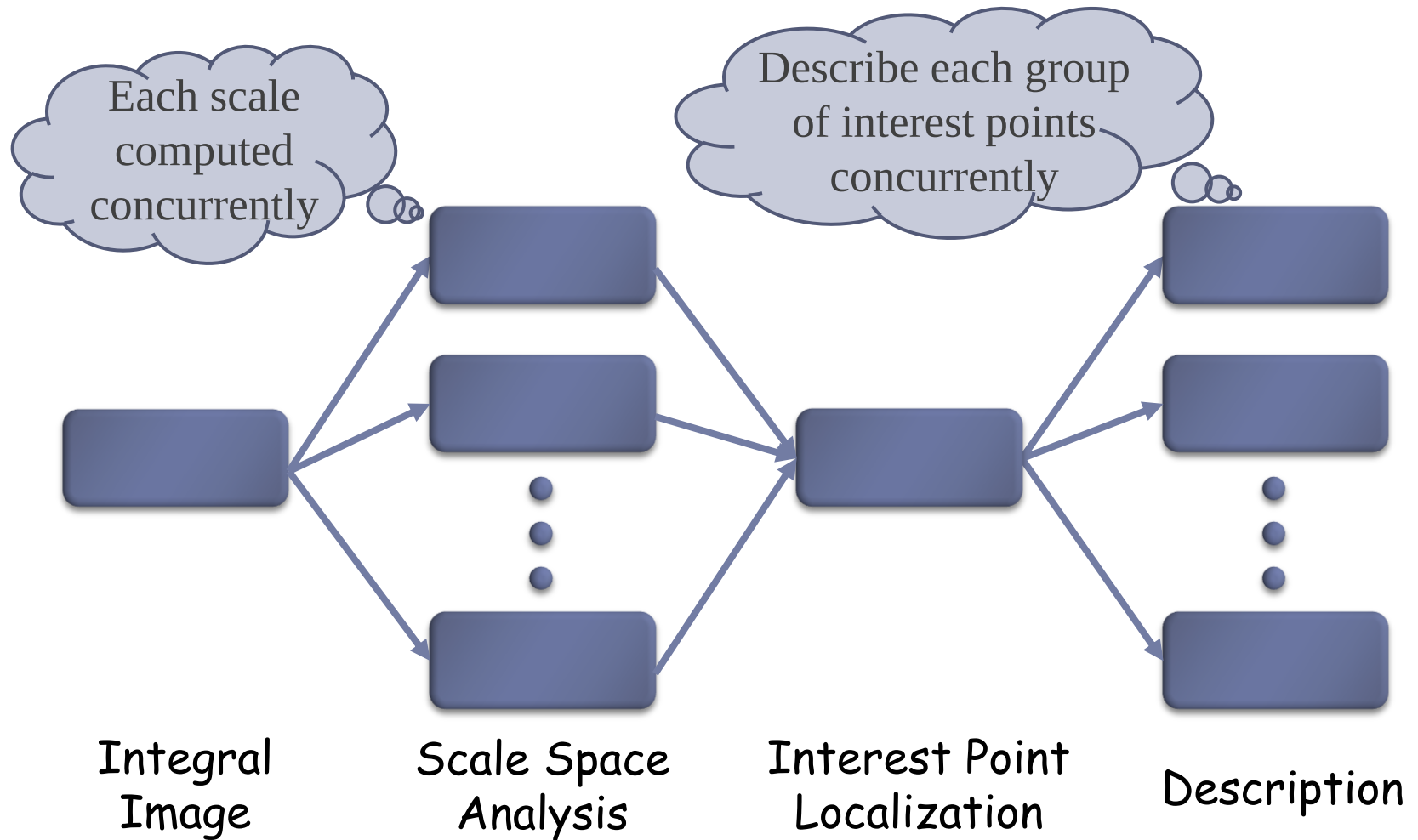
Task Parallelism

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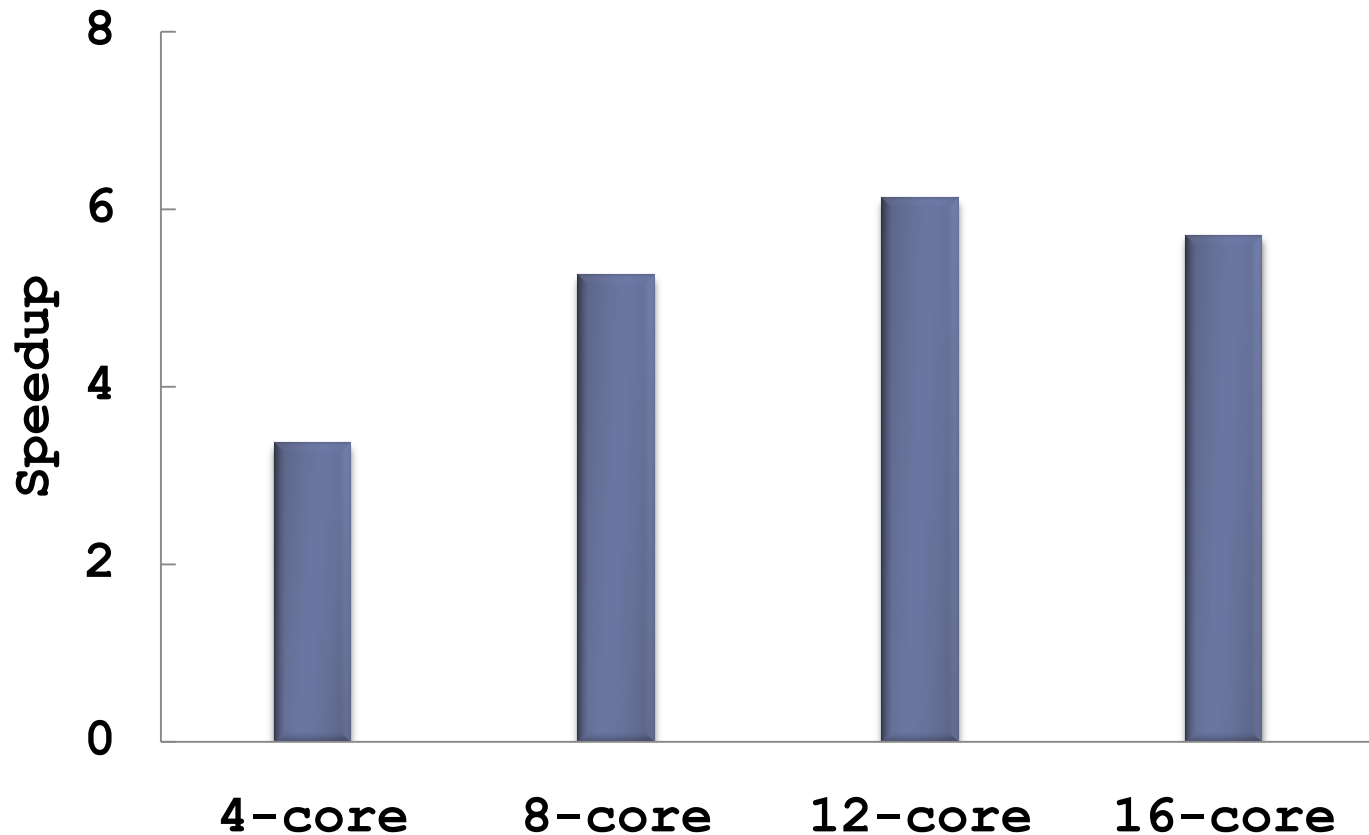
- ▶ Combination of SIMD to Others
- ▶ Combination of Task and Pipeline Parallelism

Scale-level Parallelism



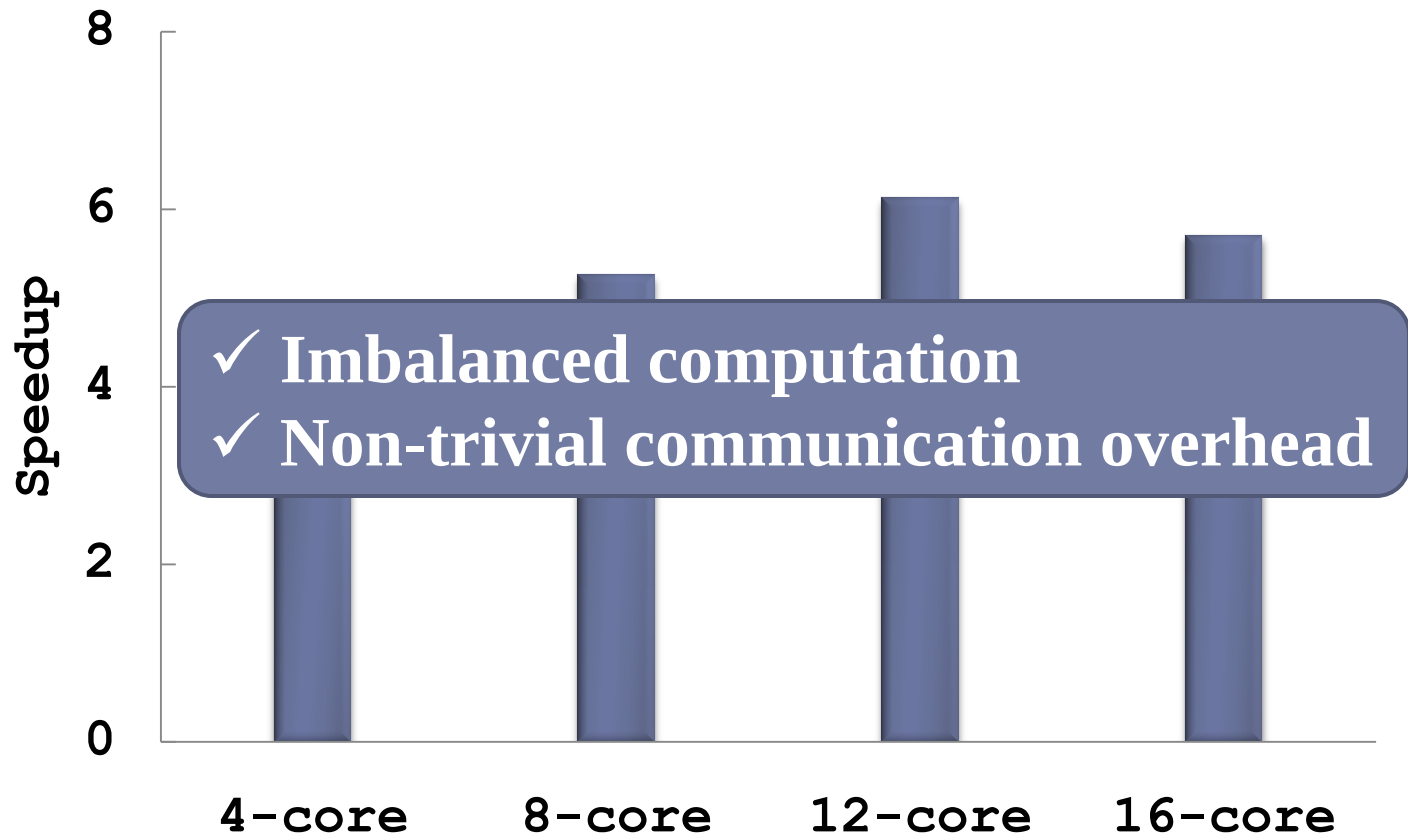
Results of Scale-level Parallelism

Not scale when exceeding 12 cores



Results of Scale-level Parallelism

Not scale when exceeding 12 cores



Parallel Analysis

Pipeline Parallelism

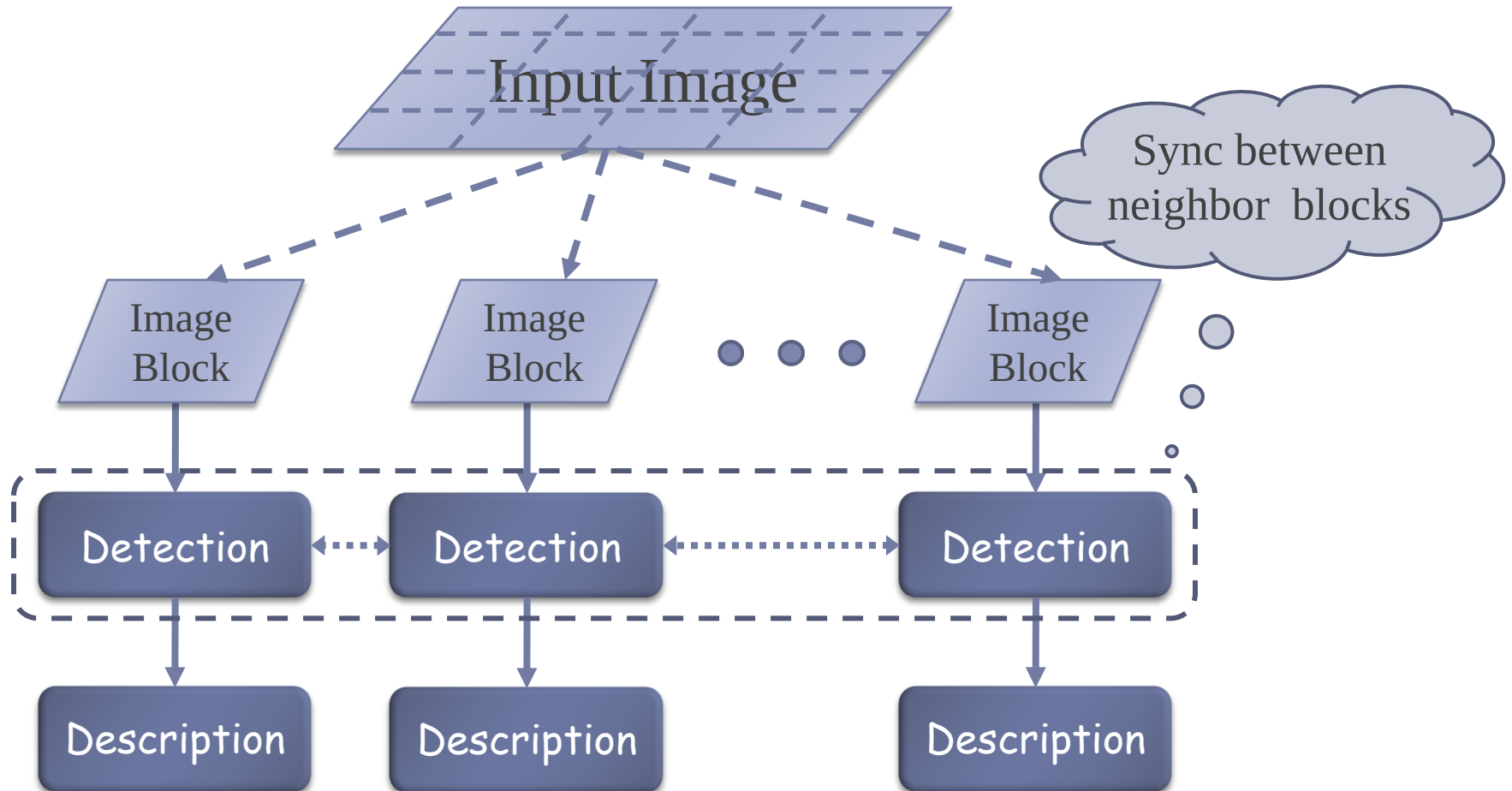
Task Parallelism

- ▶ Scale-level Parallelism
- ▶ Block-level Parallelism

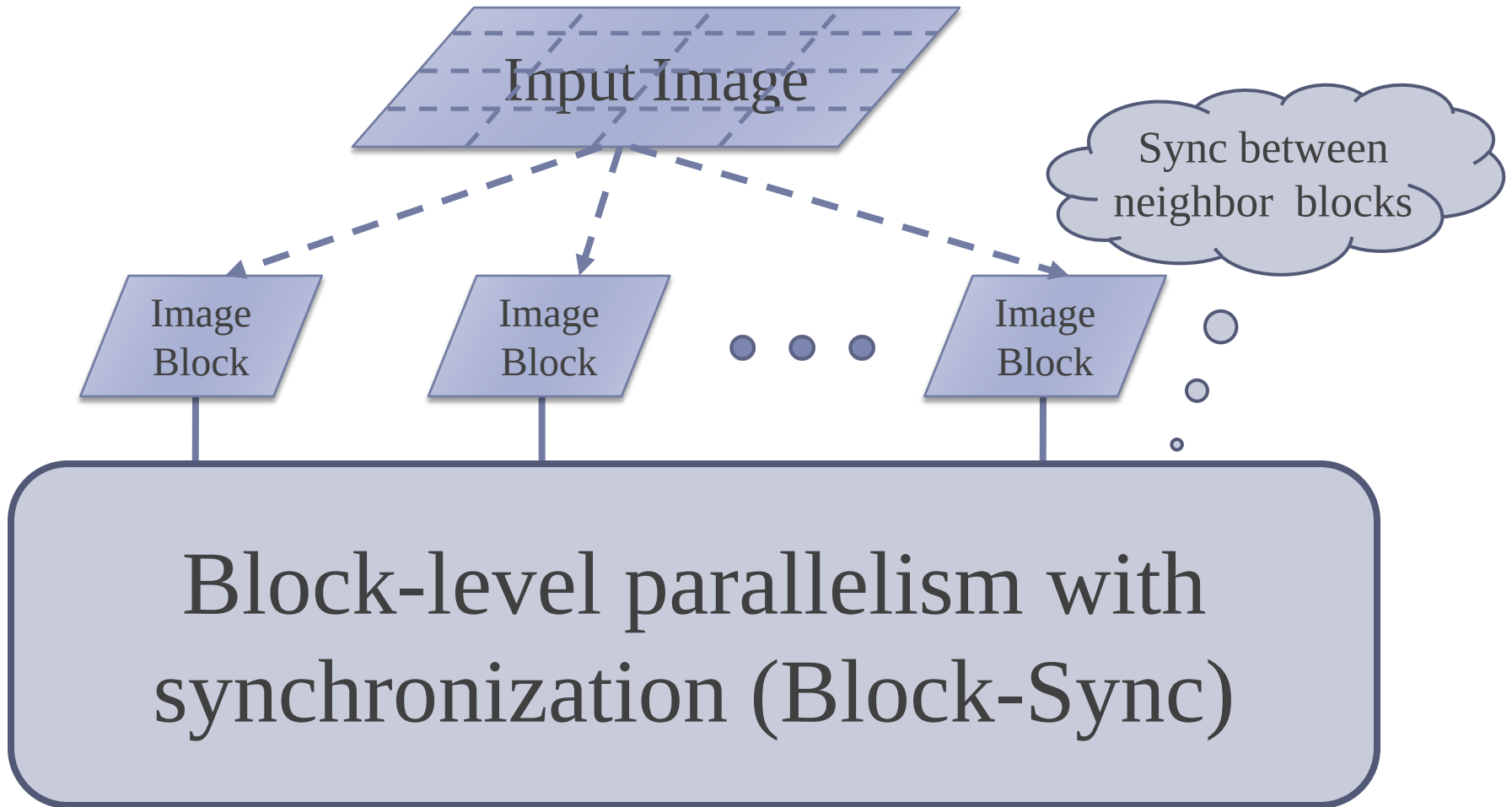
Combination of Different Parallelism

- ▶ Combination of SIMD to Others
- ▶ Combination of Task and Pipeline Parallelism

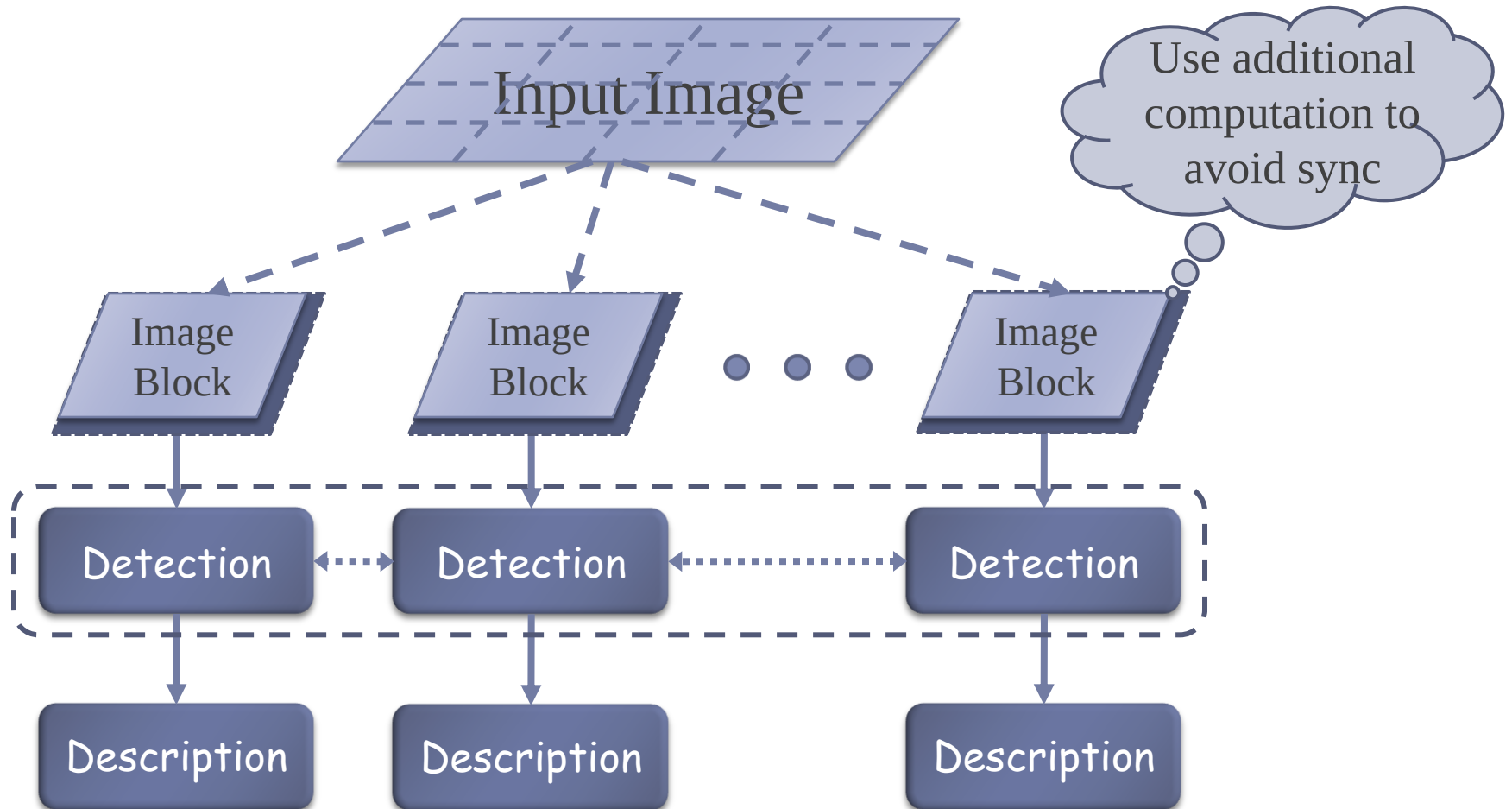
Block-level Parallelism



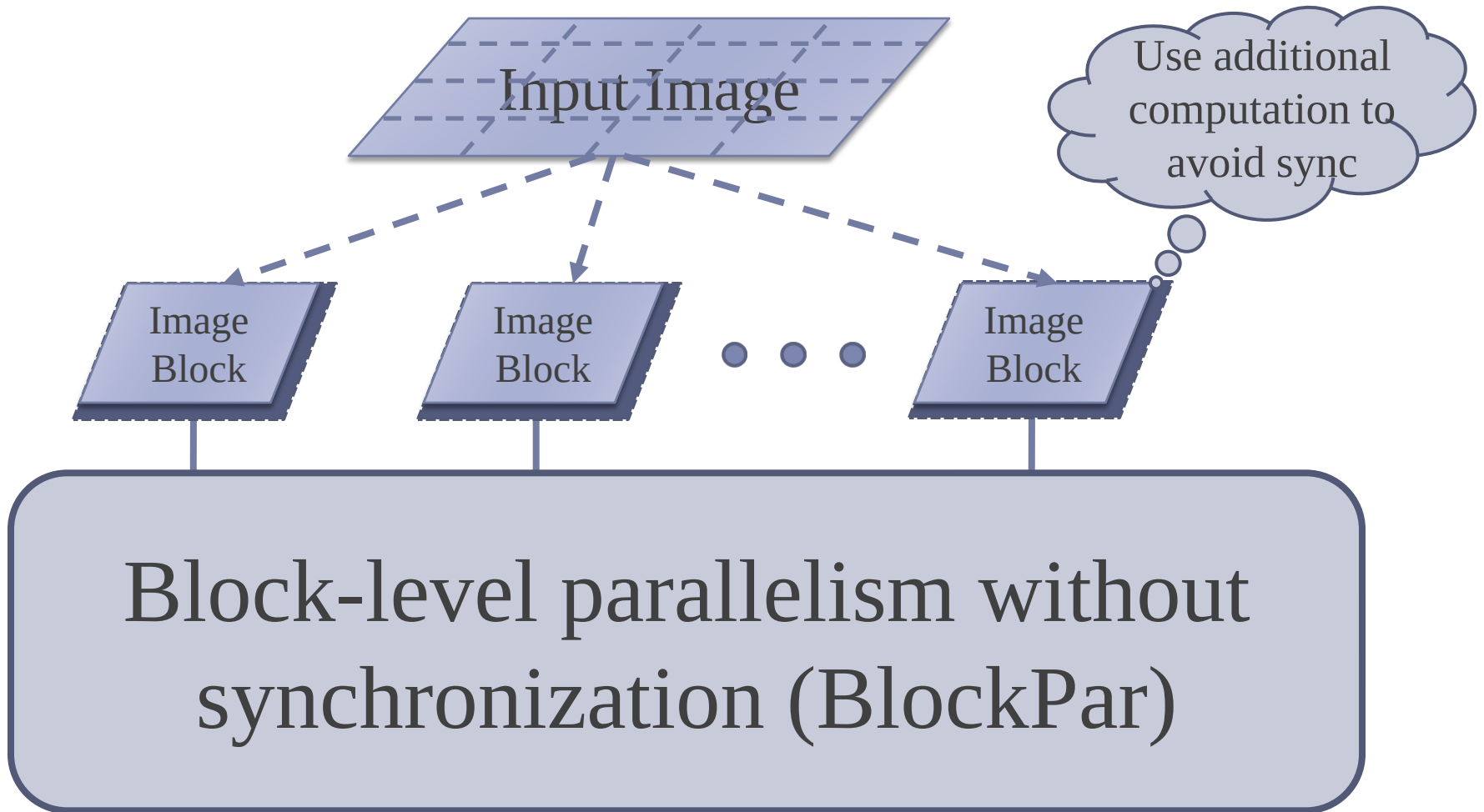
Block-level Parallelism



Block-level Parallelism

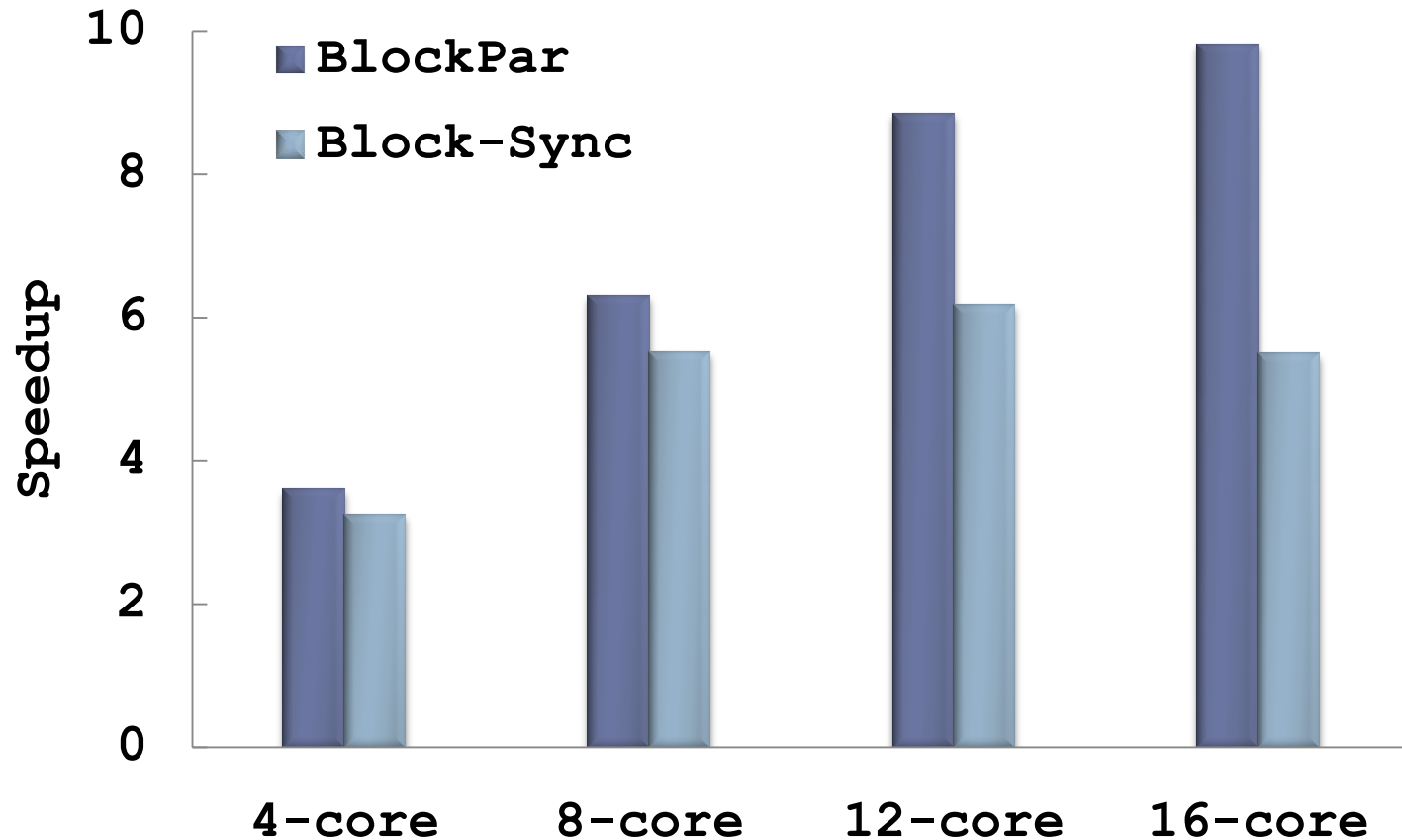


Block-level Parallelism



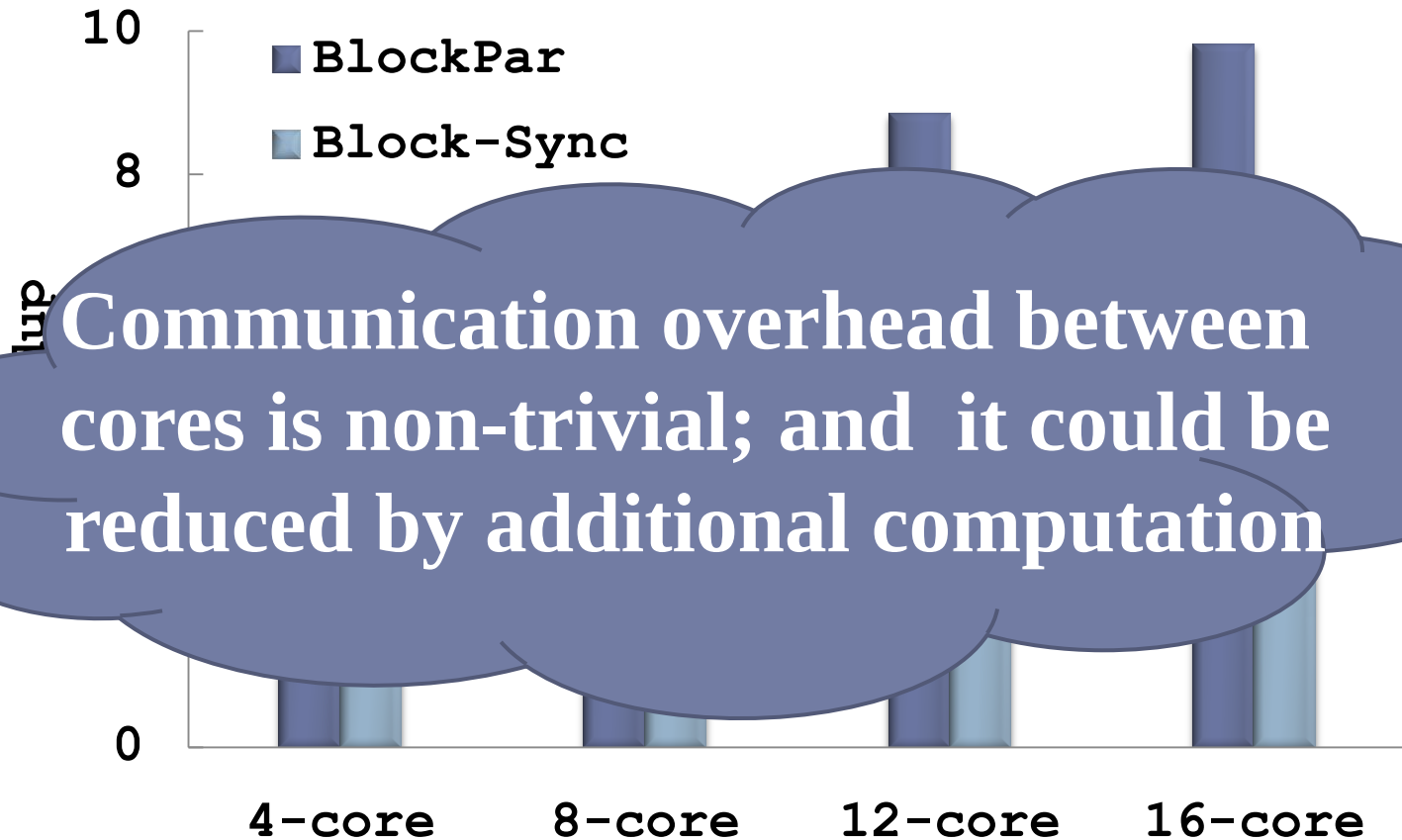
Results of Block-level Parallelism

BlockPar scales well



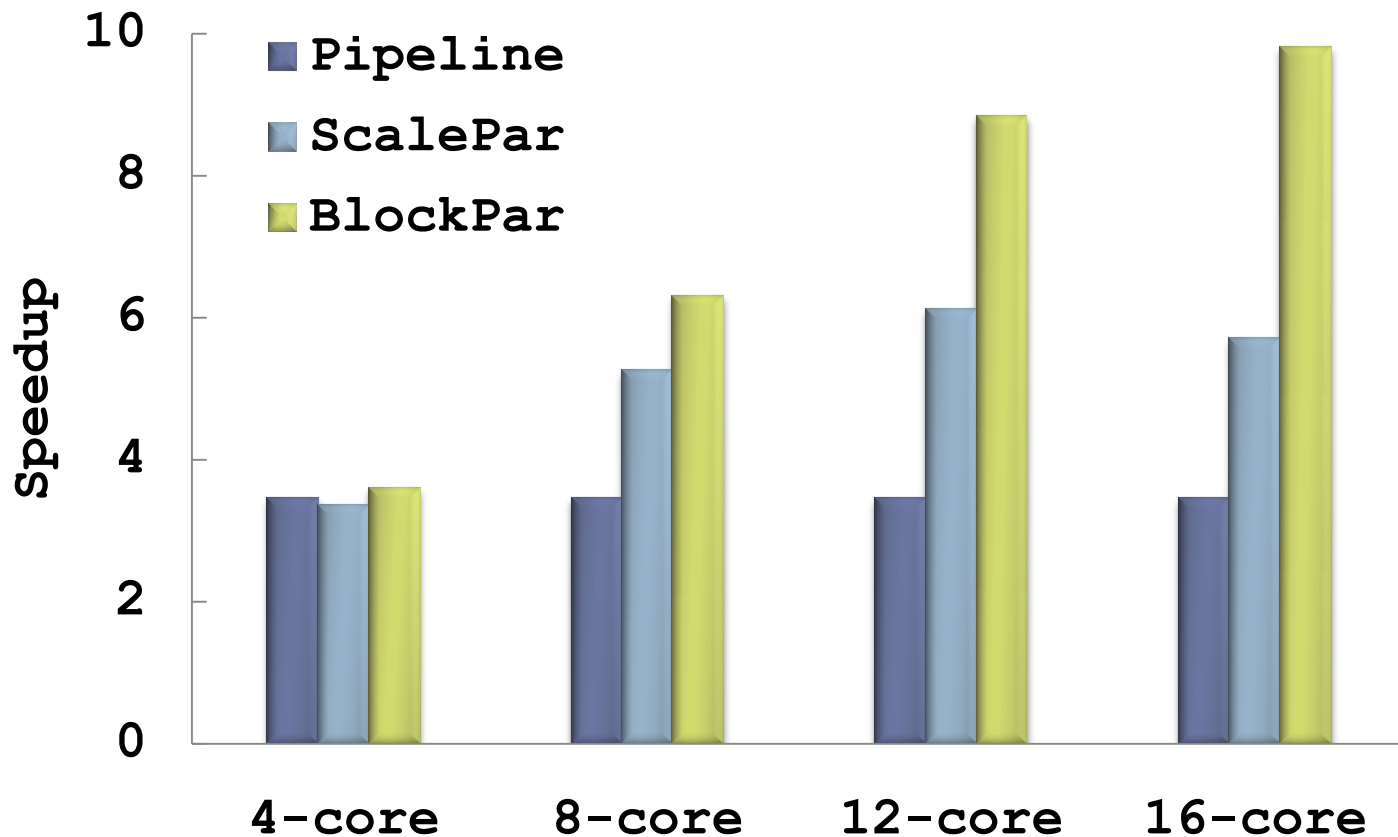
Results of Block-level Parallelism

BlockPar scales well



Comparison for Each Parallelism

Block-level parallelism is more efficient



Parallel Analysis

Pipeline Parallelism

Task Parallelism

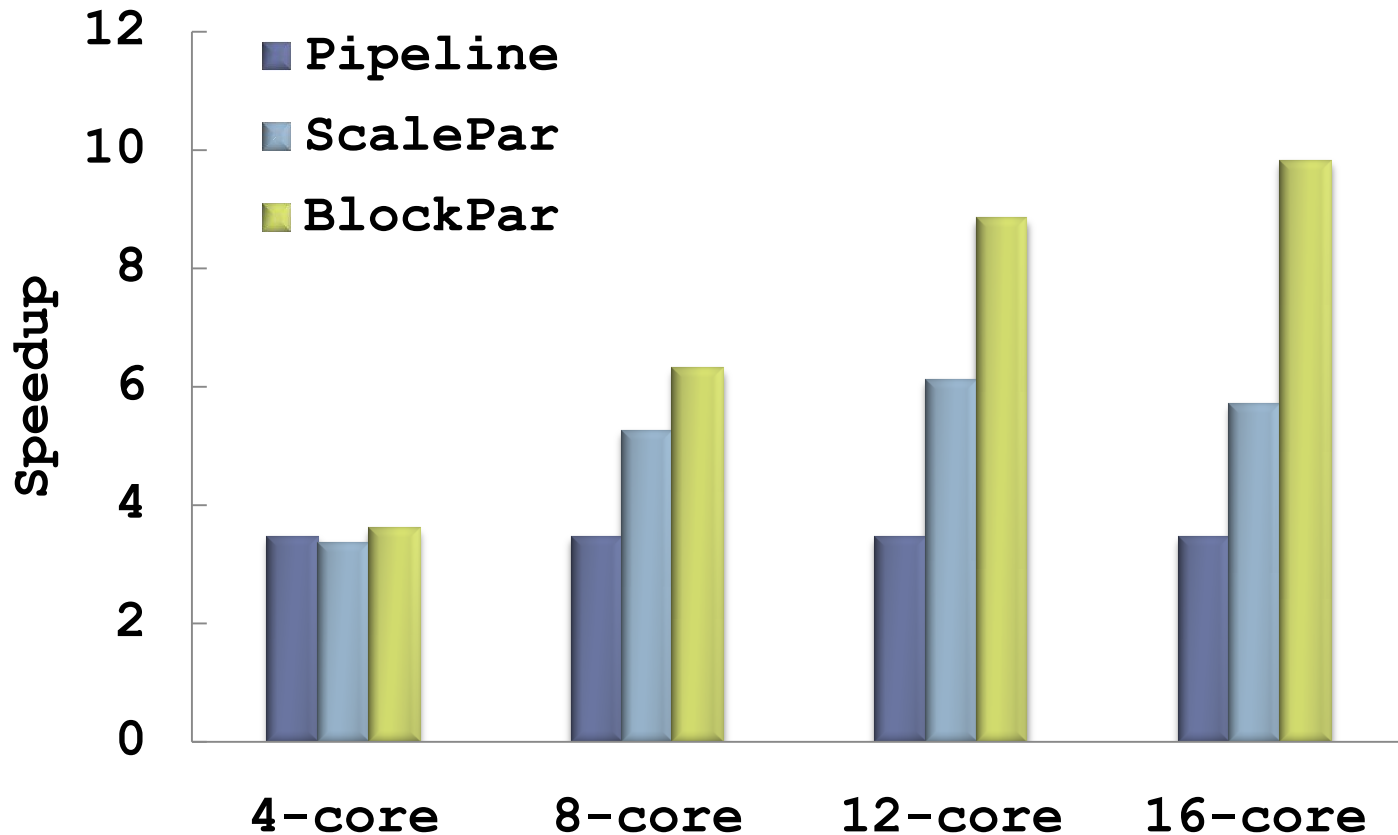
- ▶ Scale-level Parallelism
- ▶ Block-level Parallelism

Combination of Different Parallelism

- ▶ Combination of SIMD to Others
- ▶ Combination of Task and Pipeline Parallelism

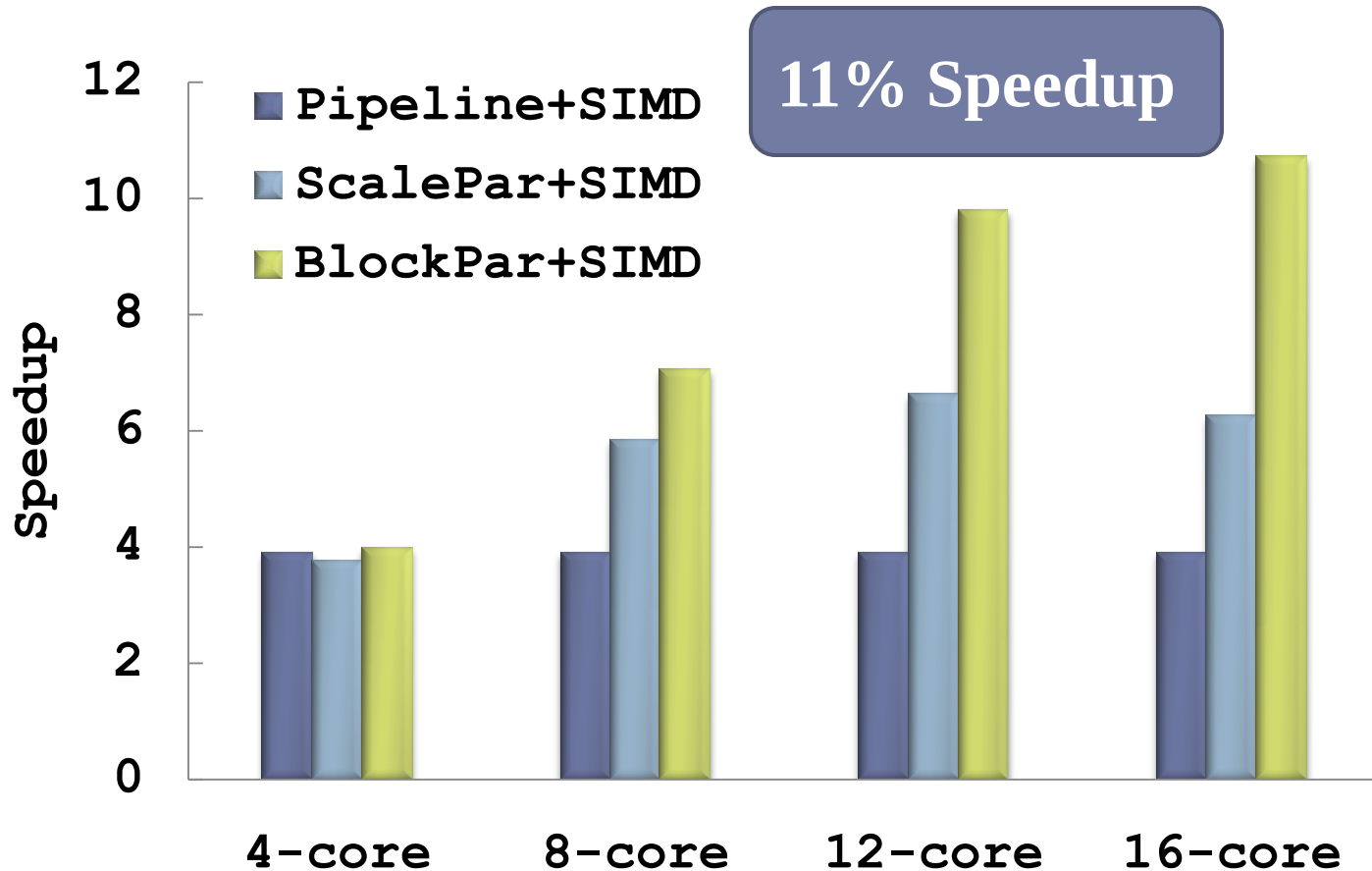
Combination of SIMD to Others

Use ICC to generate SIMD instructions



Combination of SIMD to Others

Use ICC to generate SIMD instructions



Parallel Analysis

Pipeline Parallelism

Task Parallelism

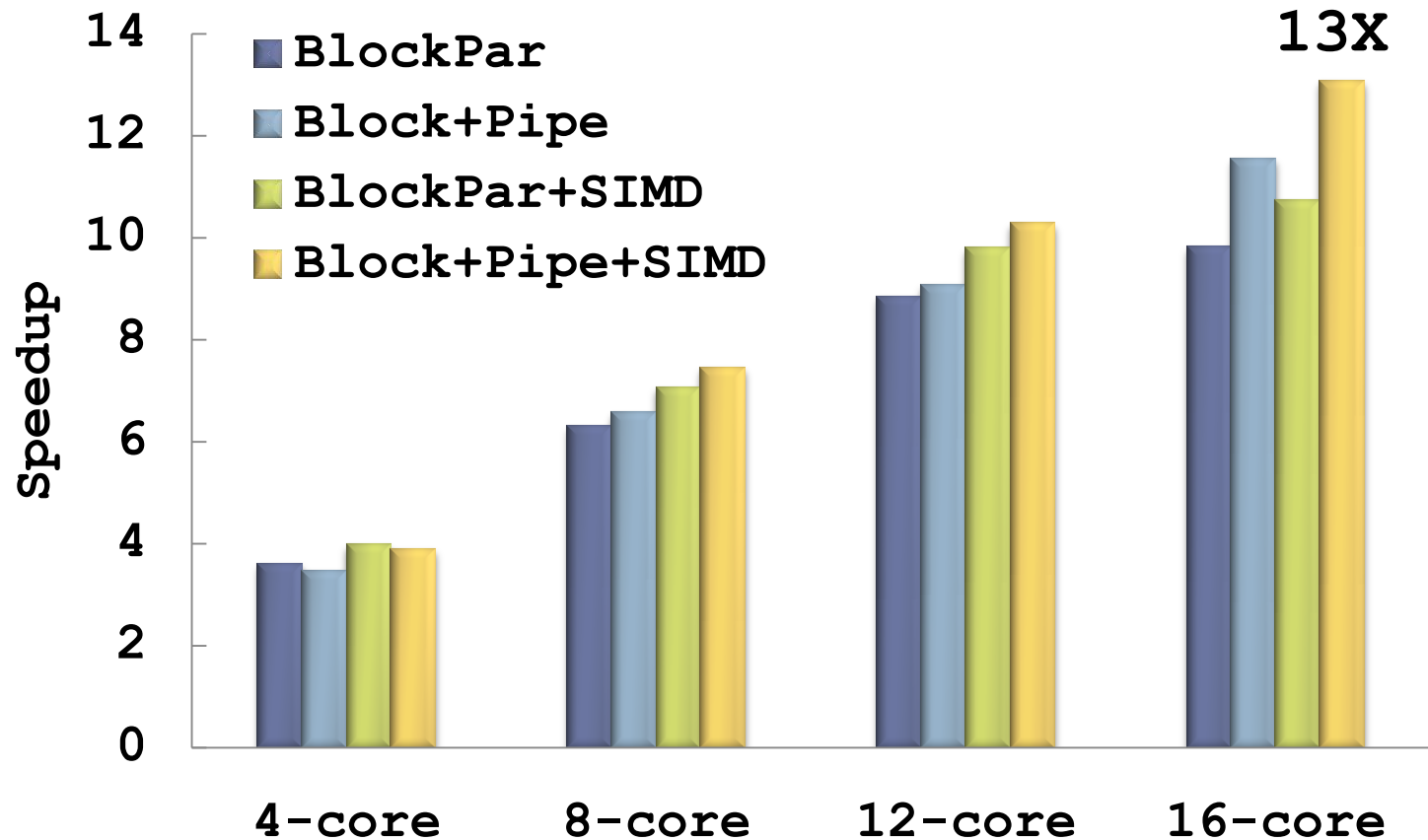
- ▶ Scale-level Parallelism
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- ▶ Combination of Task and Pipeline Parallelism

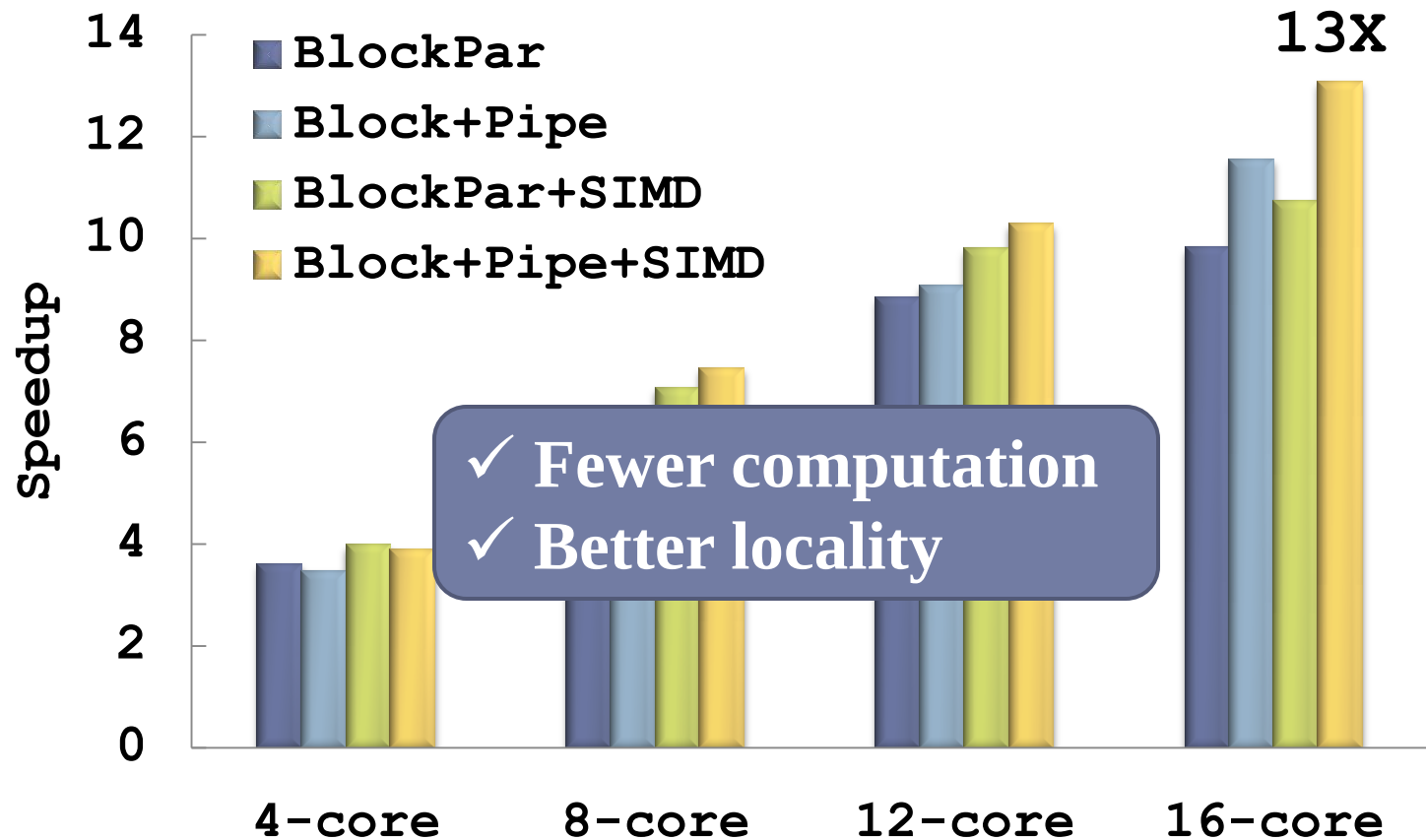
Combination of Task & Pipeline

BlockPar + Pipeline is the most efficient



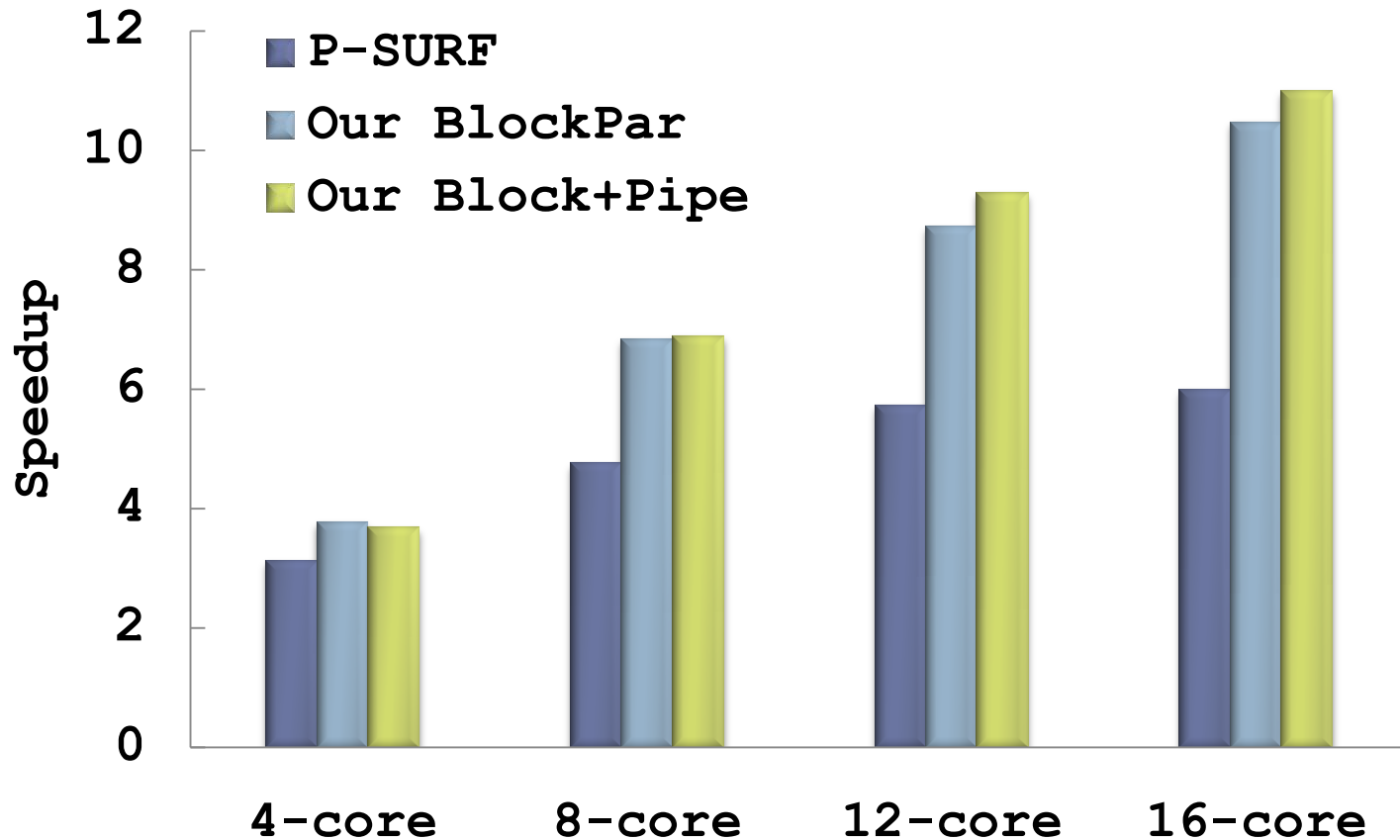
Combination of Task & Pipeline

BlockPar + Pipeline is the most efficient



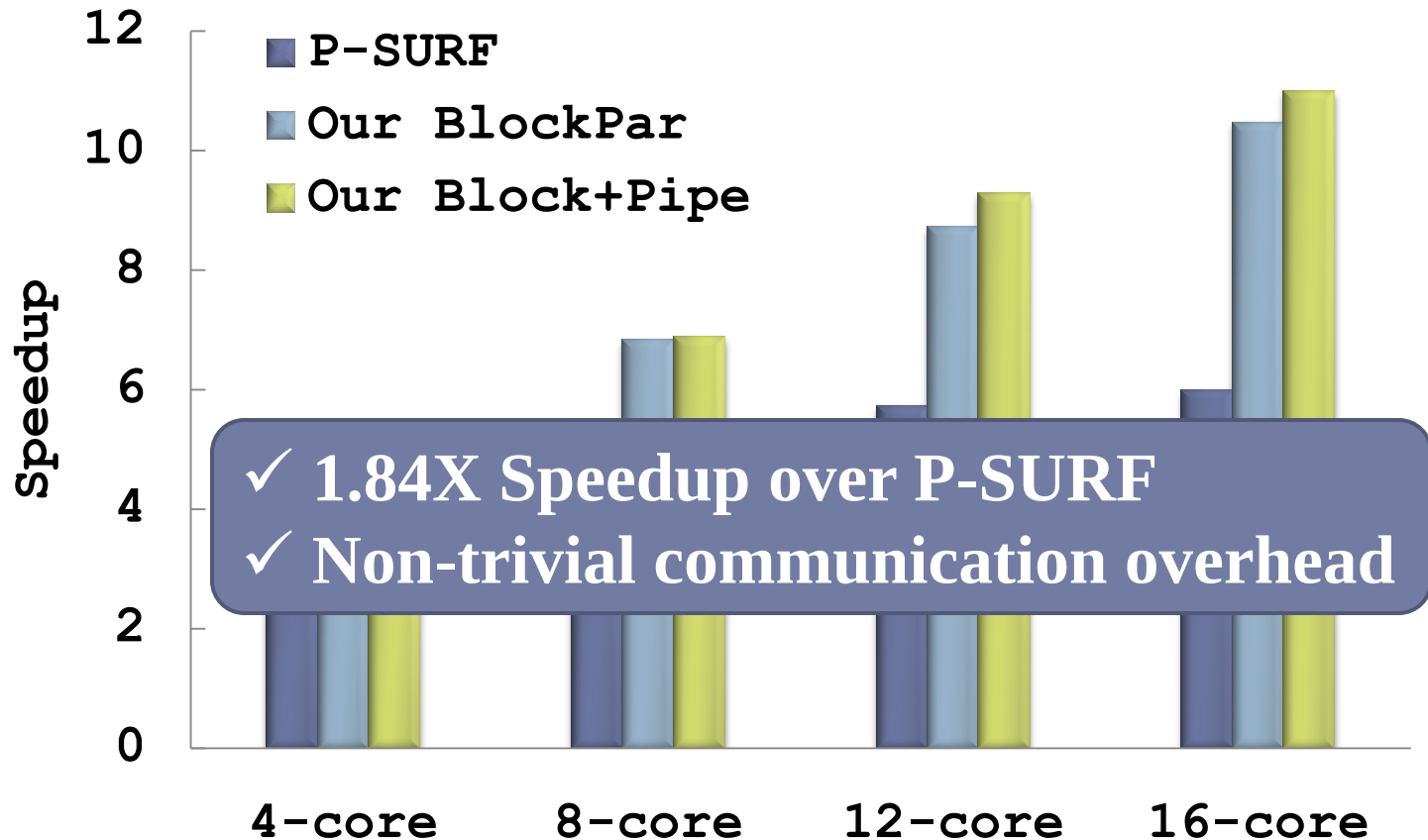
Comparison to Prior Work

Compared to *P-SURF* [Zhang 10] on *multi-core CPU*



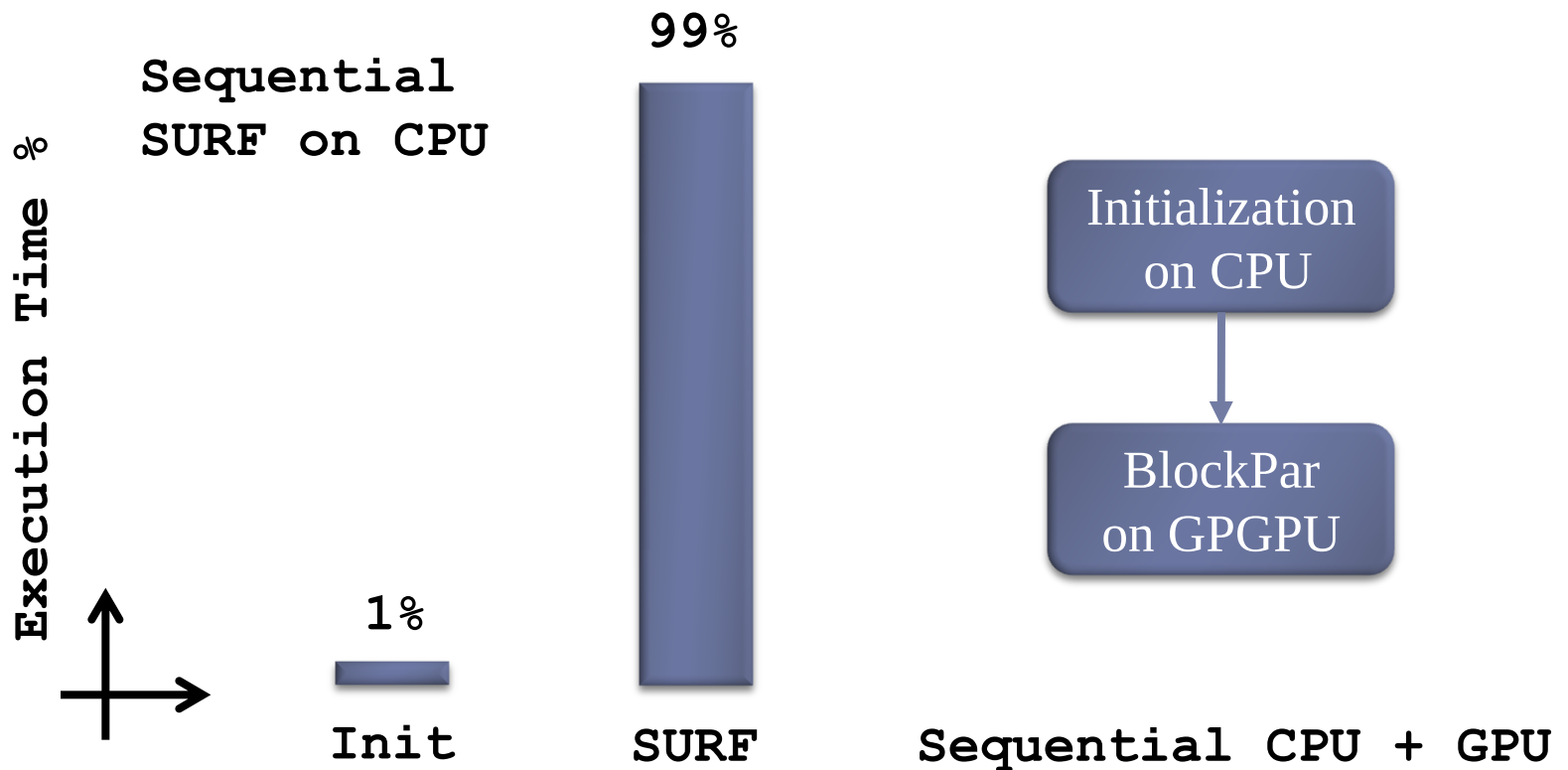
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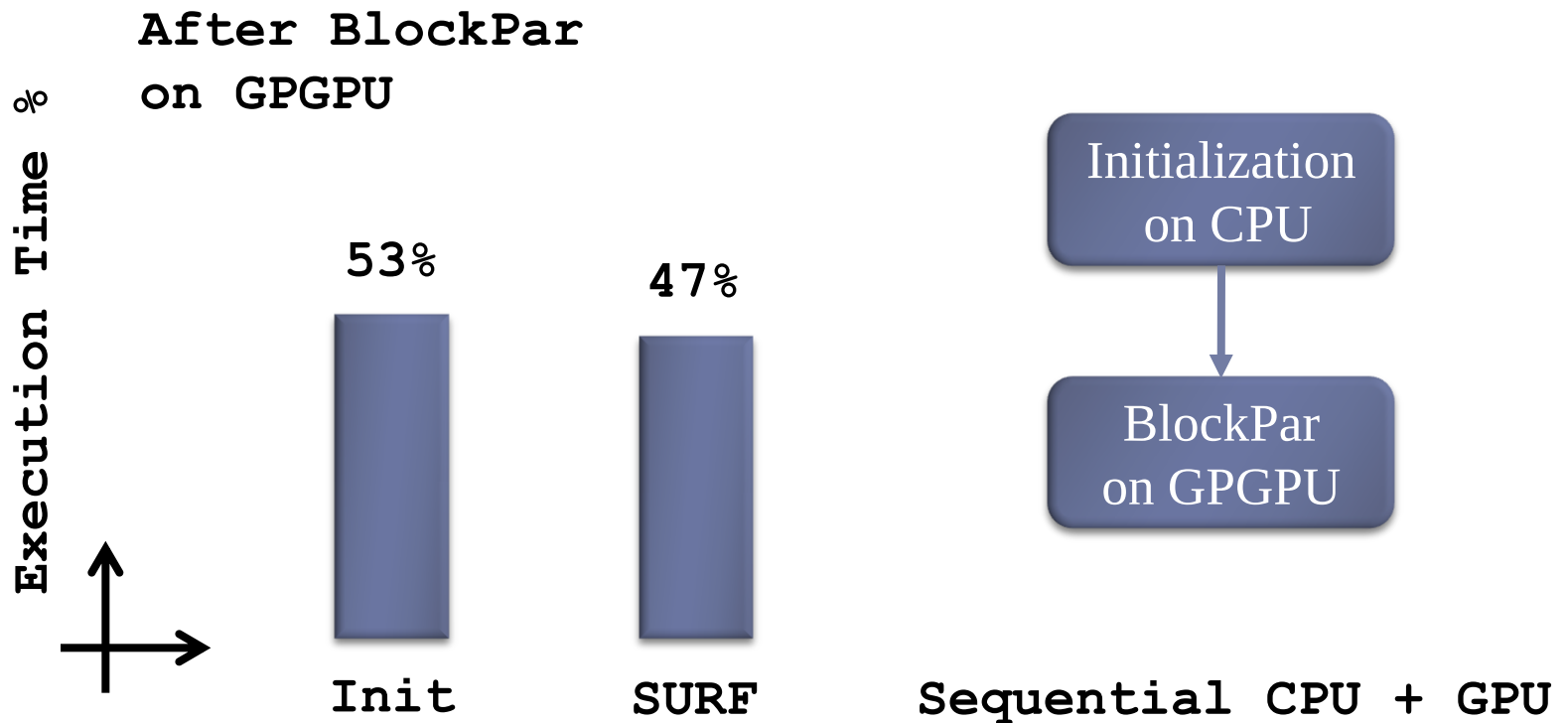
Comparison to Prior Work (cont.)

Our implementation on **GPGPU**



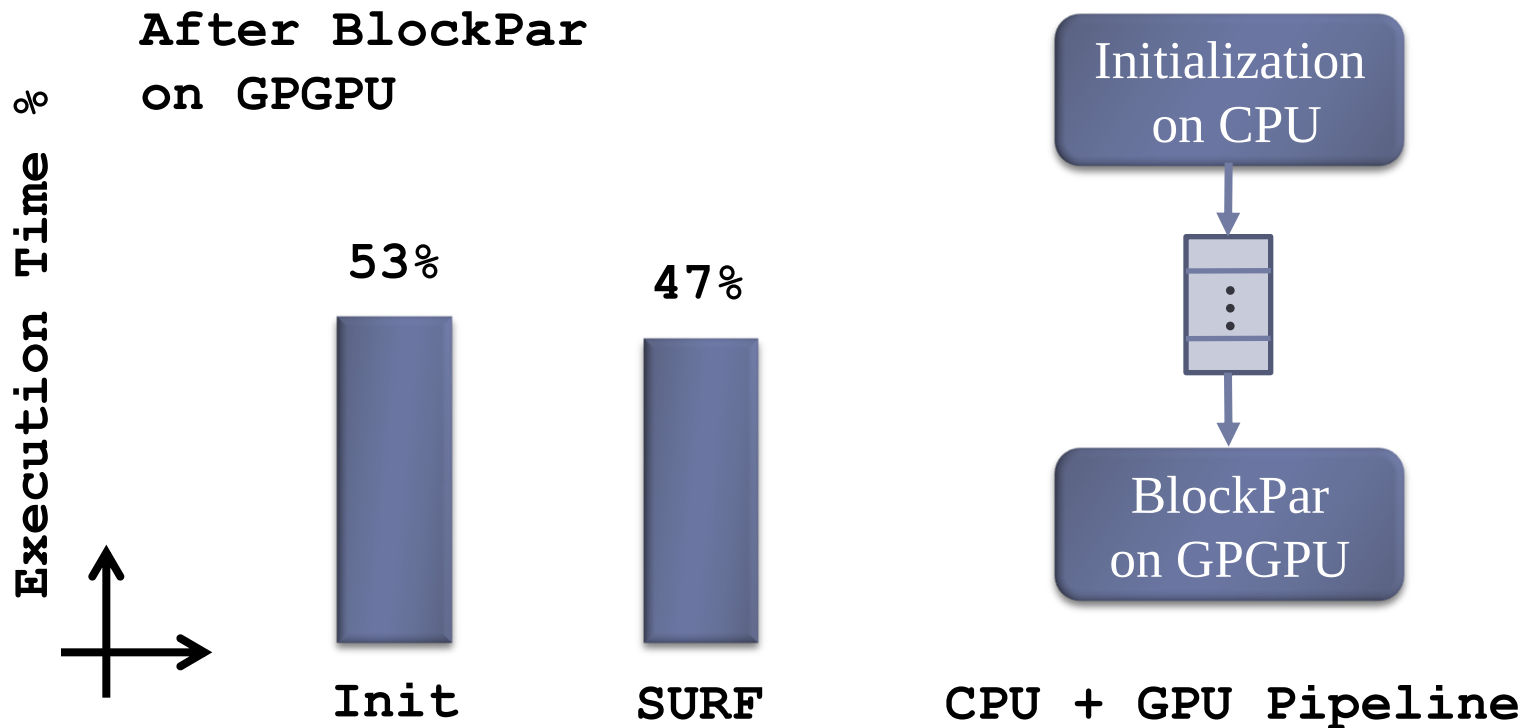
Comparison to Prior Work (cont.)

Our implementation on **GPGPU**



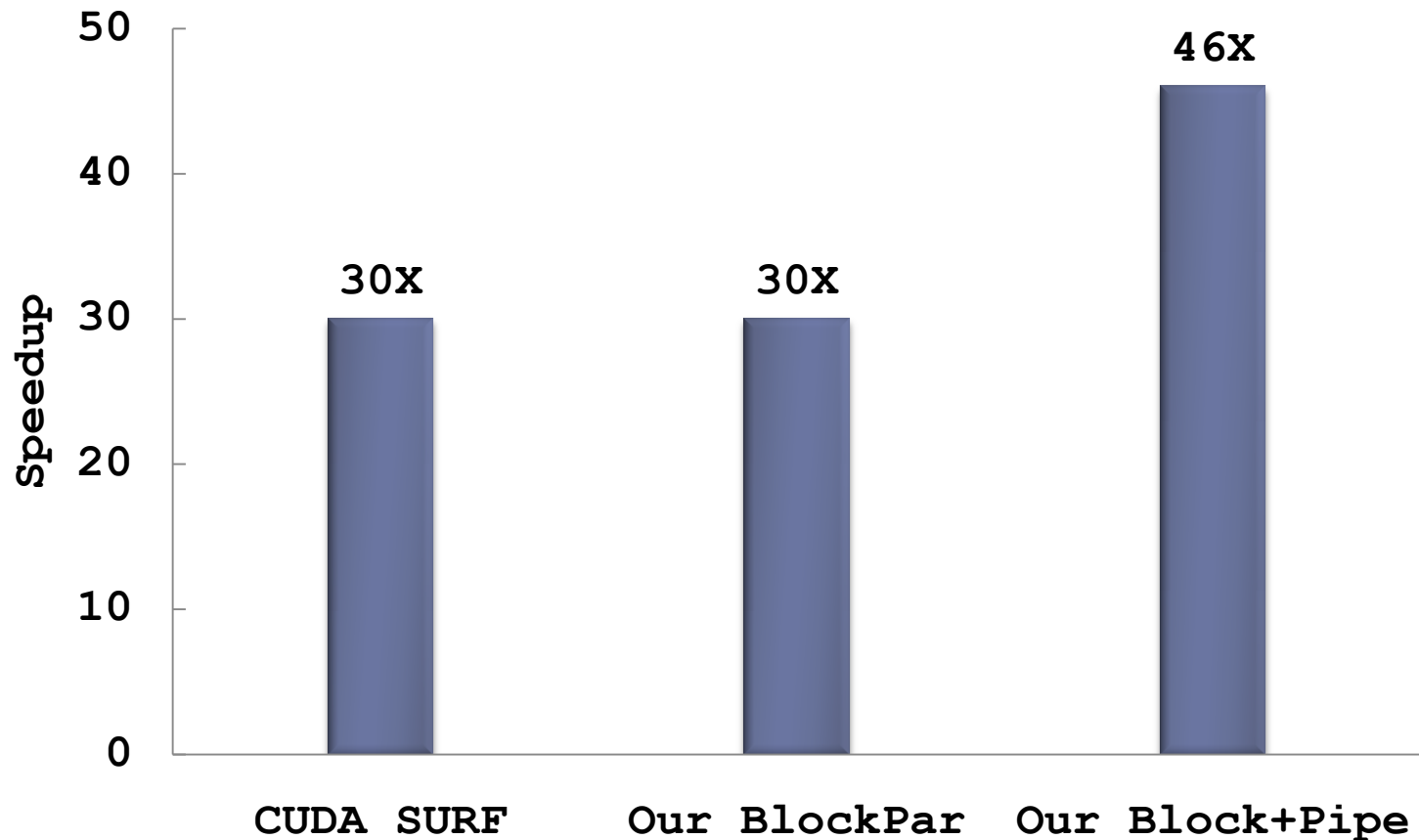
Comparison to Prior Work (cont.)

Our implementation on **GPGPU**



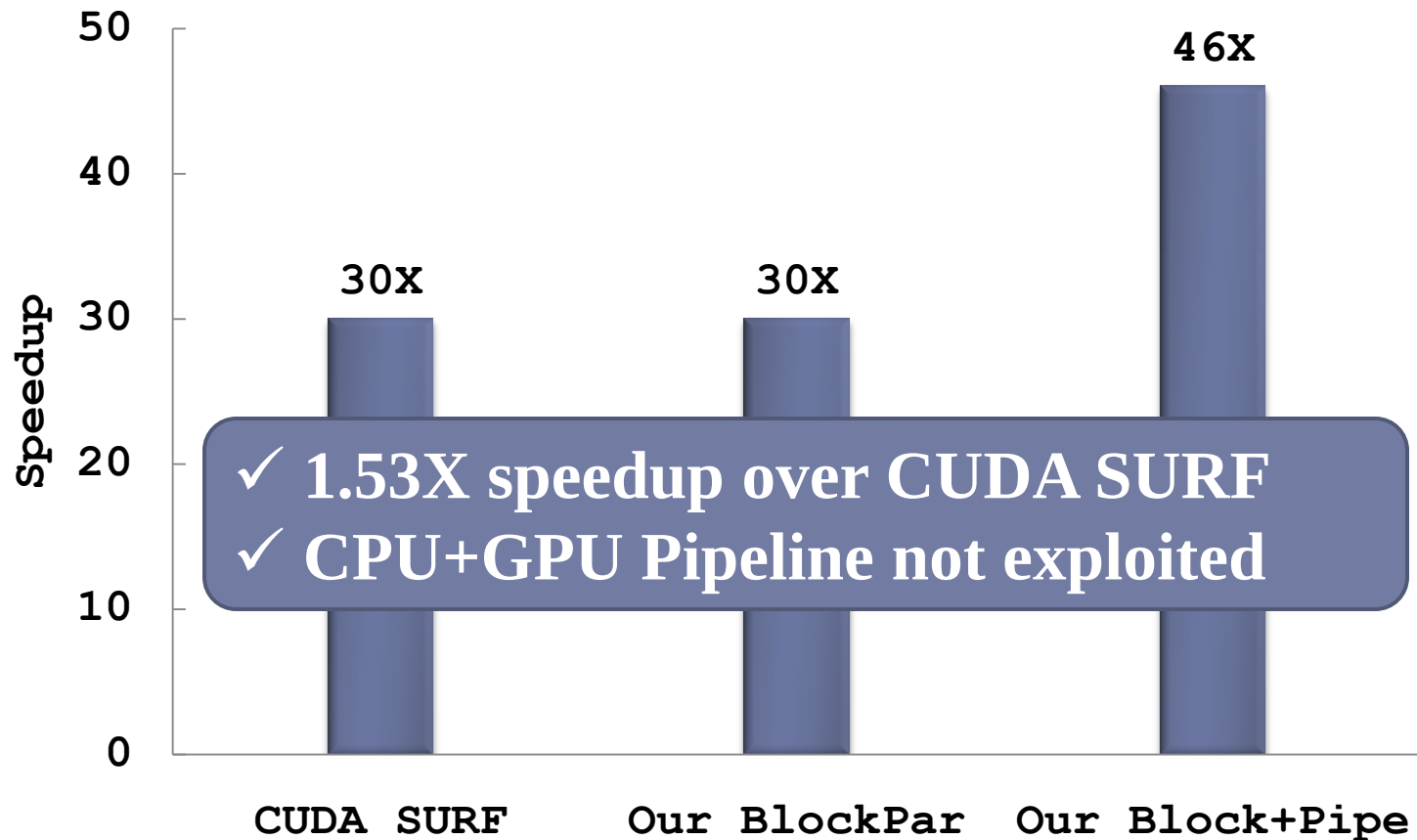
Comparison to Prior Work (cont.)

Compared to **CUDA SURF*** on **GPGPU** (*Nvidia GTX 260*)



Comparison to Prior Work (cont.)

Compared to **CUDA SURF*** on **GPGPU** (*Nvidia GTX 260*)



Summary

First parallelism analysis of image retrieval

- ▶ *Pipeline Parallelism*
- ▶ *Task Parallelism at scale-level & block-level*
- ▶ *Data Parallelism, i.e., SIMD*
- ▶ *Their combinations*

BlockPar + Pipeline is the most efficient

- ▶ *13X speedup on 16-core CPU, 1.84X faster than P-SURF*
- ▶ *46X speedup on GPU, 1.53X faster than **CUDA SURF***

Conclusion and Future Work

Additional computation to avoid synchronization

Cooperation between CPU & GPU

Future work

- ▶ Apply parallel analysis to speech recognition
- ▶ Design some energy-efficient architecture, such as FPGA, to accelerate multimedia retrieval

Thanks



Parallel Processing Institute

<http://ppi.fudan.edu.cn>

